# MARITIME COMMAND

# QUALIFICATION STANDARD & PLAN

## Marine Engineering Technician 313

## Naval Machinist Course

## Journeyman Developmental Period

## Qualification Code: AHHD

## MITE ID: 109033

## RECORD OF AMENDMENTS

The Authority reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of Maritime Staff DMTE. The Curriculum Control Authority for the training is CFFSE. The latest approval date entries from Chief of Maritime Staff and the Training Establishment (CCA) should appear in the bottom right corner of every page.

| REVISION | Approval date | Authority |
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## FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff and the Commandant of Canadian Forces Fleet School Esquimalt. It was prepared in accordance with:

a. A-P9-000 Series, Canadian Forces Manual of Individual Training,
b. A-PD-055-003/PQ-001 Occupation Specialty Specification OSS AHHD

2. This publication is effective on receipt and supersedes QSP-MARENG-NM 22 JUN 94. Performance Objectives (POs) are listed in the Table of Contents and are explained fully in Chapter 2 (note that the PO numbers maintain the 100 series to remain congruent with past references and Chapter 4 (EOs). The Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainees must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to

summarize all requested amendments in writing to the applicable approving authority(ies). Final approval for amendments are as follows:

a. Chapters 1 and 2 - Chief Maritime Staff, DMTE. Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment (Canadian Forces Fleet School Esquimalt); and

b. Chapters 3 and 4 - Commandant of Canadian Forces Fleet School Esquimalt. Training Establishments that conduct training and are not the CCA must forward suggested amendments to Canadian Forces Fleet School Esquimalt for approval.

4. The designated CCA(s) for this training is indicated below:

| CCA | Phase | Qualification |
|---|---|---|
| CFFSE | | AHHD |

# CHAPTER1: TRAINING MANAGEMENT DETAILS

**AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/DMTE**

AIM

1. The aim of this course is to train precision machining skills required of senior machinists in HMC Ships and Canadian Forces Naval Training Establishments. Upon completion of the Performance Objectives, the machinist will have an advanced understanding of machine capabilities, and confidence in their own abilities to machine metal stock or parts in HMC Ships and duties as a machine shop worker.

COURSE DURATION CALCULATION

2. Course length is calculated in accordance with MARCORD 9-30.

| | |
|---|---|
| Actual Training Days (Total Hours Of Instruction Ã· 6hrs / Training Day | 109 |
| Admin Day (if required) | 9 |
| Travel Time (if required) | |
| Course Duration (Total) | 118 |

RESOURCES

3. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H.

| **DEDICATED COURSE STAFF** | | | |
|---|---|---|---|
| **POSITION** | **RANK** | **SPECIALTY** | **NO.** |
| Instructor | PO1 | Provincially qualified machinist, and/or;<br>Naval Machinist specialist course (AHHD qualification) graduate having been recommended as an instructor. | 1 |
| | | | |

| Rooms Dedicated to the Course (Classroom, Trainer, Theatre) | | Number of Days in Dedicated Area |
|---|---|---|
| Bldg # | Room Number / Facility | |
| 92 | Machine Shop | 109 |

COURSE CAPACITY

4. Planning for the conduct of training is based on the following guidelines:
   a. maximum - 6
   b. minimum - 4
   c. optimal - 6

COURSE PREREQUISITES

5. The candidate must:

   be a MAR ENG Tech 00122 QL5A graduate;

   achieve 80%(minimum) during QL5 machine shop phase; or

   if criteria b is not met, a waiver may be granted following the completion of a machine shop project to an 80% standard in a specified time, and receive permission in writing from the commandant CFFSE/CFNES.

RELATED DOCUMENTS

6. The related document is OSS AHHD.

EVALUATION

7. This course will be evaluated through administered written and practical Performance Checks, where the trainee must achieve a minimum of 75% for each PC.

VALIDATION

8. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in annex B.

QUALIFICATION

9. Successful completion of this course constitutes the primary means of attaining the Naval Machinist qualification Code AHHD (MITE course ID 109033).

PRIOR LEARNING ASSESSMENT

10. Prior Learning Assessment shall be conducted in accordance with A-P9-050-000/PT-012 Canadian Forces Military Equivalencies Program and

MARCORD 9-47.

SPECIAL INSTRUCTIONS

11. N/A

# CHAPTER 2: PERFORMANCE OBJECTIVES

**AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY CHIEF MARITIME STAFF/ DMTE**

PO 001

1. Performance. Manage a Machine Shop

2. Conditions.

   a. Given:

      (1) operational machine shop; and

      (2) applicable references

   b. Denied:

      (1) n/a

   c. Environmental:

      (1) equipped naval workshop at sea or alongside.

3. Standard.

   a. IAW applicable references manage a machine shop to include:

      (1) enforce shop safety;

      (2) evaluate machinery;

      (3) maintain machine shop inventory;

      (4) interpret machining procedure specifications; and

      (5) classifying metals

4. References.

   A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

   1, 2, 3, 4

6. Supporting Skills.

   1, 3, 5, 15, 27

7. Supporting Knowledge.

1, 2, 3, 4, 5, 8, 38, 39, 41, 42, 53, 98, 104

8. Training Limitations.
None Identified

PO 002

1. Performance. Perform Machine Shop Fitting

2. Conditions.

a. Given:

(1) operational machine shop;

(2) hand tools and equipment ; and

(3) applicable references

b. Denied:

(1) n/a

c. Environmental:

(1) equipped naval workshop at sea or alongside.

3. Standard.

a. IAW a working drawing, use fitting techniques and work within specified/required tolerances.

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

5, 6, 7

6. Supporting Skills.

4, 7, 11, 12, 13

7. Supporting Knowledge.

6, 17, 18, 24, 26, 34, 36, 37, 74, 99, 100, 101, 103

8. Training Limitations.

None Identified

PO 003

1. Performance. Perform Advanced Grinding Operations

2. Conditions.

a. Given:

(1) Operational machine shop;

(2) Grinder and accessories;

(3) Special tools and equipment; and

(4) Applicable references.

b. Denied:

(1) n/a

c. Environmental:

(1) Equipped naval workshop at sea or alongside.

3. Standard.

a. Demonstrate precision grinding procedures IAW references.

b. Carry out grinding stone:

(1) Identification and classification;

(2) Truing and dressing;

(3) Mounting; and

(4) Inspection.

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

8

6. Supporting Skills.

16, 17, 18, 19,

7. Supporting Knowledge.

43, 44, 45, 46, 47, 48, 49, 50, 51, 52

8. Training Limitations.

None Identified

PO 004

1. Performance. Perform Precision Drill Press Operations

2. Conditions.

a. Given:

(1) Operational machine shop;

(2) Drill press and accessories;

(3) Special tools and equipment; and

(4) Applicable references.

b. Denied:

(1) n/a

c. Environmental:

(1) Equipped naval workshop at sea or alongside.

3. Standard.

a. IAW references:

(1) Prepare work piece for drilling; and

(2) Operate a drill press and accessories to specified tolerances of work required.

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

10

6. Supporting Skills.

30

7. Supporting Knowledge.

9, 77, 81, 82, 86, 93

8. Training Limitations.

None Identified

PO 005

1. Performance. Perform Advanced Machine Lathe Operations

2. Conditions.

a. Given:

(1) operational machine shop;

(2) machine lathe and accessories;

(3) special tools and equipment; and

(4) applicable references.

b. Denied:

(1) n/a.

c. Environmental:

(1) equipped naval workshop at sea or alongside

3. Standard.

a. IAW references including working drawings/specifications:

(1) calculate spindle speeds and feed rates;

(2) align the tailstock;

(3) prepare lathe attachments for use;

(4) perform advanced lathe operations; and

(5) machine lathe work to specified/required tolerance.

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

9

6. Supporting Skills.

20, 22, 24, 25, 26, 31,

7. Supporting Knowledge.

14, 54, 55, 58, 59, 60, 61, 62, 63, 64, 65, 66, 69, 70, 71, 72, 73, 76, 79, 80, 87, 102,

8. Training Limitations.

None Identified

PO 006

1. Performance. Perform Advanced Milling Machine Operations

2. Conditions.

a. Given:

(1) Operational machine shop;

(2) Milling machine and accessories;

(3) Special tools and equipment; and

(4) Applicable references.

b. Denied:

(1) N/A

c. Environmental:

(1) Equipped naval workshop at sea or alongside

3. Standard.

a. IAW references to include working drawings/specifications:

(1) Install milling machine attachments;

(2) Mount milling and specialty cutters; and

(3) Perform advanced milling machine operations to specified/required tolerances.

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

11

6. Supporting Skills.

21, 23, 29

7. Supporting Knowledge.

10, 11, 40, 88, 89, 90, 91, 92, 94, 95, 96, 97

8. Training Limitations.
None Identified

PO 007

1. Performance. Machine Components

2. Conditions.

a. Given:

(1) Operational machine shop; and

(2) Applicable references.

b. Denied:

(1) Assistance

c. Environmental:

(1) Equipped naval workshop at sea or alongside

3. Standard.

a. Machine parts IAW drawing specifications

4. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

5. Specification Task Numbers.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

6. Supporting Skills.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31

7. Supporting Knowledge.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 102, 103, 104.

8. Training Limitations.

None Identified

# CHAPTER 3: ASSESSMENT OF TRAINEE

## AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF CANADIAN FORCES FLEET SCHOOL ESQUIMALT (CFFSE)

GENERAL

1. Performance Checks (PCs), based on the Performance Objectives (POs) in Chapter 4 will be the basis for trainee assessment. A trainee must pass all PCs or the supplementary PC to successfully complete the Naval Machinist Course and receive Naval Machinist Qualification AHHD.

2. Pass/Fail assessment of each candidate is based on the successful completion of all Performance Objectives as stated in Chapter 4 of this publication. Performance Checks consist of a written examination and practical projects. The naval machinist course an elite course that offers advanced training to select QL5 graduates. The nature of the job demands excellence so that machined parts perform the intended function under operational conditions, and meet the restrictions of shipboard facilities as well as time. Therefore, each candidate must attain a minimum of 75% for each of the following PCs to successfully complete the Naval Machinist Course:

Performance Checks:

a. PC 002 Perform Machine Shop Fitting: Practical Project (Triangle and Plate);
b. PC 004 Perform Precision Drill Press Operations: Written exam to include TPs covered in PO 001, PO 003 and PO 004;
c. PC 005 Perform Advanced Machine Lathe Operations: Practical Project (Mini Bomb;
d. PC 006 Perform Advanced Milling Machine Operations: Practical Project (V block and clamp); and
e. PC 007 Machines Components: Practical Projects (Part A - Drill Press, Part B - Gear Pump and Part C - Pump shaft.

PERFORMANCE CHECKS

3. Trainees will be permitted a maximum of three component restarts for each Performance Check (PC). The candidate will be penalized five per cent from the project mark for each restart. Projects to be completed within the timeline specified regardless of restarts. More than three restarts will constitute a failure of the PC.

4. Due to time and resource requirements to carry out the machining projects, supplementary performance checks cannot be given. If the candidate fails after three

restarts, a Training Review Board has the authority to recommend the performance check be repeated, in part, if time permits.

5. Performance Checks will be administered in the form of written examinations and practical machining projects. A minimum mark of 75 per cent for each Performance Check is required for successful completion of the Naval Machinist Course. The cumulative score of the following PC's, weighted as follows, will constitute the final mark:

| Performance Checks | Final Mark (%) |
|---|---|
| PC 002 Perform Machine Shop Fitting……... | 5% |
| Practical project: Triangle and Plate | |
| PC 004 Perform Precision Drill Press Operations…… | 5% |
| Written exam to include TPs covered in PO 001, PO 003 and PO 004 | |
| PC 005 Perform Advanced Machine Lathe Operations……..... | 10% |
| Practical project: Mini Bomb | |
| PC 006 Perform Advanced Milling Machine Operations……..… | 10% |
| Practical project: V block and clamp; | |
| PC 007 Machines Components……..…… | 70% |
| Practical projects: Part A - Drill Press (40%), | |
| Part B - Gear Pump (30%) | |
| Part C - Pump Shaft projects (30%) | |
| Total…… | 100% |

PROGRESS MONITORING

6. Continuous monitoring of candidate progress is required on the Naval Machinist course to provide the following:

a. notice of candidate difficulties;
b. information for training review boards;
c. indication of appropriate difficulty level; and
d. indication of progress in specific machine operation skill development.

7. Instructor's Divisional Notes:

a. Instructor's Divisional Notes will be used to record all observations on candidate performance. They will include:

(1) applying job knowledge and skills;
(2) motivation towards work;
(3) response to direction;
(4) initiative;
(5) problem solving;
(6) performance under stress;
(7) learning;
(8) effect on group morale;
(9) communication.

8. Following the completion of each Performance Objective, the trainee will complete a feedback questionnaire on the material, teaching and practical aspects of the PO. These questionnaires will provide the instructor and Training Establishment with candidate feedback on learning difficulties, presentation of course material, ways to improve training, and any problems that may affect the learning process for future courses.

UNSATISFACTORY TRAINING PROGRESS

9. The training review process is contained in MARCORD 9-60 Volume 1. Unsatisfactory course progress is indicated by the failure of a Performance Check (not an Enabling Check). A Training Review Board must then decide on the seriousness of the failed PC and recommend a course of action to the Commandant of the training establishments (CFFSE/CFNES Halifax).

10. Due to time and resource requirements to complete the machining projects, supplementary performance checks cannot be given. However, trainee's are allowed a maximum of three restarts for each performance check

TRAINING REVIEW BOARD

11. All Training Review Board requirements are contained in MARCORD 9-60 Volume 1.

COURSE REPORTS

12. Canadian Forces Course Report (form CF-377) will be submitted for each trainee. The preparation and distribution of form CF-377 shall be in accordance with DAOD

5031-9.

# CHAPTER 4: ENABLING OBJECTIVES

## AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT OF CANADIAN FORCES FLEET SCHOOL ESQUIMALT

**PO 001 Manage Machine Shop**

**EO 001.01**

1. Performance. Enforce Machine Shop Safety

2. Conditions.

   a. Given:

      (1) Operational machine shop, and

      (2) References.

   b. Denied: n/a

   c. Environmental:

   (1) equipped machine shop; and

   (2) classroom.classroom.

3. Standard.

   a. Explain shop safety regulations and procedures; and

   b. Identify unsafe practices and procedures.

4. Time.

   a. 5 X 50 minute periods for instruction

   b. 1 X 50 minute period for EC

5. Method/Media.

   a. Instruction

   b. Multimedia.

6. References.

   A1, A2, A3

7. Training Aids.

   a. Multimedia: Video # 38657 6301

8. Test Details.

a. EC: This EC will consist of a written exam administered by instructor on completion of this EO to confirm TPs covered in para 10;

b. Pass mark is 75%;

c. 1 x 50 minute period for EC; and

d. No re-write will be given to trainee failures, a Trainee Experiencing Difficulty form is to be completed and the Trainee is to be strictly monitored.

9. Remarks.

a. All requirements for WHMIS training as described in EO 001.01 para 10 must be met, prior to exposure to any controlled product; and

b. Written EC administered by instructor is to be distributed from standards

10. Teaching Points.

a. Shop safety requirements IAW References to include:

(1) Personal Protective Equipment (PPE);

(2) Safety rules for general machine operation;

(3) Safety rules for lathes;

(4) Safety rules for milling machines;

(5) Safety rules for drilling machines;

(6) Safety rules for grinding machines;

(7) Safety rules for hand tools;

(8) Safe lifting procedures; and

(9) Housekeeping.

b. Explain and describe for all Controlled Products listed in para 3 the following IAW requirements of Canada Labour Code PT II and associated regulations including A-GG-040-004/GG-001 and DNDP 40 Vol. 4 Hazmat Safety:

(1) Product Identifier or Hazardous Substance;

(2) Hazardous Information disclosed on the MSDS and Control Label; and

(3) Handling, Storage and Emergency Procedures to include but not limited to the following:

(a) Dustbane sweeping compound;

(b) Lapping/grinding compound;

(c) Steel ink (Aerosol and Liquid);

(d) Steel ink remover;

(e) Anti seize;

(f) Vision and eye cleaner;

(g) Hand cleaner;

(h) Methanol absolute;

(i) Aluminum cutting fluid; and

(j) Lathe cutting fluids.

## Enabling Check EO 001.01 – Machine Shop Safety

**PART I - ADMINISTRATIVE INSTRUCTIONS**

MATERIAL AND FACILITIES REQUIRED

1. Given a Quiz
2. Classroom

**ARRANGEMENTS AND PROCEDURES**

1. Instructor to arrange a classroom through Senior Instructor.
2. Instructor to review TPs covered in EO 001.01 prior to administering EC.
3. No re-write will be given for trainee failures, but a Trainee Experiencing Difficulty form is to be completed and the Trainee is to be strictly monitored.
4. This EC will not be weighted towards final course mark.
5. Standards to issue EC
6. Instructor to review EC 24hrs prior to administering

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. 1 x 50 minute period for EC
2. Pass mark is 75%

**PO 001 Manage Machine Shop**

**EO 001.02**

1. Performance. Interpret Machining Procedure Specifications

2. Conditions.

   a. Given:

      (1) References;

      (2) Machining specifications;

      (3) Operational machine shop;

      (4) Maintenance schedules;

      (5) Assistance, and

      (6) Supervision.

   b. Denied: n/a

   c. Environmental:

      (1) Equipped machine shop.

3. Standard.

   a. Identify working parts of shop equipment; and

   b. Perform machine shop calculations.

4. Time.

   a. 2 x 50 minute periods for instruction

5. Method/Media.

   a. Lecture; and

   b. Skills Practice.

6. References.

   C4, C5, D1, D3

7. Training Aids.

a. Specifications drawings;

b. Shop machinery; and

c. Calculator.

8. Test Details.

a. The TPs covered in EO 001.02 will be assessed as PC 004 on completion of PO 004 as a written exam.

9. Remarks.

a. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points.

a. Identify working parts of shop equipment to include:

(1) Lathes;

(2) Grinders;

(3) Millers;

(4) Drillers;

(5) Cutters;

(6) Hand tools;

(7) Power saws;

(8) Hydraulic press; and

(9) Power tools.

b. Skills Practice: Perform machine shop calculations

(1) Interpreting Machining formula;

(2) Using machining charts and tables;

(3) Spindle speed calculation (drill press, lathe, milling machine.)

c. Skills Practice: Interpreting machining procedures and specifications

(1) Setups,

(2) Time appreciation,

(3) Layouts machine,

(4) Techniques shop capabilities,

(5) Resources, and

(6) Interpret technical drawings

**PO 001 Manage Machine Shop**

**EO 001.03**

1. Performance. Maintain Machine Shop Inventory

2. Conditions.

   a. Given:

      (1) Operational machine shop and hand tools;

      (2) Applicable stock;

      (3) Test equipment;

      (4) References;

      (5) Supervision; and

      (6) Assistance.

   b. Denied: N/A

   c. Environmental:

      (1) Operational machine shop

3. Standard.

   a. IAW references:

      (1) Identify metals;

      (2) Describe heat treatment processes of metals;

      (3) Select cutting fluids;

      (4) Store tools; and

      (5) State material storage considerations.

4. Time.

   a. 24 x 50 min periods for instruction

   b. 16 X 50 minute periods for the foundry

5. Method/Media.

a. Instruction;

b. Demonstration;

c. Videos;

d. Skills Practice; and

e. Homework assignments.

6. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1

7. Training Aids.

a. VHS Video's;

b. Student Handouts;

c. Texts/Manuals;

d. Material samples;

e. Operational machine shop;

f. PPE; and

g. Operational foundry.

8. Test Details.

a. The TPs covered in EO 001.03 will be assessed as PC 004 on completion of PO 004 as a written exam.

9. Remarks.

a. Trainee to assist in the casting process, casting project components.

b. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004.

10. Teaching Points.

a. Describe physical properties of metals:

(1) definition of ductility, malleability, brittlemess, elasticity, hardness,

toughness and fatigue failure;

(2) definition of machinability;

(3) definitioin of working hardness - the hardening of material caused by repeated bending or flexing. Explanation of the annealling process to soften and remove the internal and external stresses of the metal.

b. Ferrous and non-ferrous metals:

(1) describe ferrous metals;

(2) describe non ferrous metals: explain alloys in steels and the effects of adding the elements of carbon, manganese, silicon, sulphur, phosphorus, nickel, chromium, molybdenum, vanadium and tungsten.

c. Identification of metals by appearance:

(1) visually identify metals using chart;

(2) determine the grade of steel by using spark test; and

(3) identify metals using a magnet.

d. Describe SAE and AISI classification systems.

e. Explain the Galvanic Scale and its uses to include electrode potentials and the results of galvanic coupling.

f. Describe heat treatment processes and applications to include:

(1) annealing

(2) hardening; and

(3) tempering

g. Storage and use of shop fluids to include:

(1) cutting fluids;

(2) tapping fluids;

(3) corrosion inhibitors;

(4) cleaning fluids;

(5) Prussian blue;

(6) Layout fluid;

(7) Solvents; and

(8) Lubricants.

h. Material and tool storage to include:

(1) hand tools;

(2) machine tools;

(3) metal stock; and

(4) POLs.

i. Skills Practice: Casting procedures

## PO 002 Perform Machine Shop Fitting

## EO 002.01

1. Performance. Perform Precision Measurements

2. Conditions.

   a. Given:

      (1) Precision measuring tools to include:

         (a) Vernier measuring instruments;

         (b) Micrometer measuring instruments;

         (c) Dial indicator;

         (d) Thread gauges;

         (e) Thickness gauges;

         (f) Caliper - inside and outside;

         (g) Telescopic gauges; and

         (h) Depth gauges;

      (2) References

      (3) Metal stock;

      (4) Supervision; and

      (5) Assistance.

   b. Denied: n/a

   c. Environmental:

   (1) Equipped machine shop; and

   (2) Classroom

3. Standard.

   a. IAW refs select and apply the following measuring tools:

(1) Vernier measuring instruments;

(2) Micrometer measuring instruments;

(3) Dial indicator;

(4) Thread gauges;

(5) Thickness gauges;

(6) Caliper - inside and outside;

(7) Telescopic gauges;

(8) Depth gauges; and

b. Verify the accuracy of measuring tools using a micrometer standard/gauge block.

4. Time.

a. 2 X 50 minute periods for instruction

5. Method/Media.

a. Lecture;

b. Multimedia; and

c. Skills Practice

6. References.

C5, D1,D3

7. Training Aids.

a. Precision measuring tools;

b. Videos;

c. Standards and gauge blocks; and

d. Texts and manuals.

8. Test Details.

a. The TPs covered in EO 002.01 will be confirmed on completion of EO 002.02

as practical EC 002.02 (Template Layout Project).

9. Remarks.

n/a

10. Teaching Points.

a. Describe the types, purpose, operation and care of precision instruments to include:

(1) vernier measuring instruments;

(2) micrometer measuring instruments;

(3) dial indicator;

(4) thread gauges;

(5) thickness gauges;

(6) caliper - inside and outside;

(7) telescopic gauges; and

(8) depth gauges;

b. Skills Practice: Verify the accuracy of measuring tools using a micrometer standard and gauge blocks.

## PO 002 Perform Machine Shop Fitting

## EO 002.02

1. Performance. Perform Precision Layout.

2. Conditions.

   a. Given:

      (1) Precision layout tools to include:

         (a) surface plate;

         (b) surface gauge;

         (c) hermaphrodite calipers;

         (d) dividers;

         (e) combination square;

         (f) vernier height gauge;

         (g) vernier protractor; and

         (h) metal scriber

      (2) references;

      (3) metal stock;

      (4) supervision; and

      (5) assistance.

   b. Denied: N/A

   c. Environmental:

   (1) Equipped machine shop.

3. Standard.

   a. Layout a template as per the drawing specifications IAW Ref. D3; and

   b. Observe and adhere to all machine shop safety practices.

4. Time.

a. 2 X 50 minute periods for instruction; and

b. 4 x 50 min periods for Practical EC (Layout Project).

5. Method/Media.

a. Instruction;

b. Demonstration;

c. Video; and

d. Skills Practice

6. References.

C5, D3

7. Training Aids.

a. layout tools specified in para 2.a.

b. 5" X 10", 1/8" thick mild steel (1 per student).

c. Multimedia #38657 6306.

8. Test Details.

a. Practical EC: The TPs covered in EO 002.02 will be confirmed by way of a Practical EC 002.02 (template layout project) marked by means of checklist.

b. Minimum pass mark for this EC is 75%

c. Time limit: 4 x 50 min periods for EC;

d. Assistance denied; and

e. No re-write will be given to trainee failures, a Trainee Experiencing Difficulty form will be completed.

9. Remarks.

a. Instructor is to layout tools as a display for review in para 10.

b. Material that will be used for EC template will have one side square, prepared

in advance of lesson.

c. Accuracy as specified in the working drawing

d. Instructor to supervisor trainee progress, assistance denied.

10. Teaching Points.

a. Explain types, purposes and care of layout tools to include:

(1) surface gauge;

(2) surface plate;

(3) hermaphrodite calipers;

(4) dividers;

(5) combination squares;

(6) vernier height gauge;

(7) vernier protractor;

(8) metal scriber;

(9) machinist square;

(10) angle plate;

(11) parallel bar;

(12) sine bar;

(13) punches; and

(14) hammer;

## Enabling Check EC 002.02 – Template Layout Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

### MATERIAL AND FACILITIES REQUIRED

1. Equipped machine shop;
2. Machinist tool box;
3. Precision layout tools to include:
    a. surface plate;
    b. surface gauge;
    c. hermaphrodite calipers;
    d. dividers;
    e. combination square;
    f. vernier height gauge;
    g. vernier protractor; and
    h. metal scriber
4. Working drawing complete with EC checklist;
5. Mild steel stock pre-cut to 5" X 10", 1/8" thick; 5" X 10", 1/8" thick, one side square; and
6. Shop fluids.

### ARRANGEMENTS AND PROCEDURES

1. This Enabling Check is to confirm the TPs covered in EO 002.2 by completing Template Layout Project IAW Ref D3 (working drawings and specified tolerances).
2. Assistance is not permitted.
3. Supervision is mandatory
4. Time allocated is 4 x 50 minute periods to complete EC 002.02
5. Part II lists what each trainee will be given

6. Trainees to be familiar with EC 002.02 checklist

7. Pass mark is 75% of total marks IAW EC 002.02 marking guide;

8. Each trainee will be permitted a maximum of three restarts;

9. There is a 5% penalty for restarts;

10. Failure to achieve a passing mark of 75% for EC 002.02 will result in completion of Trainee Experiencing Difficulty Form

11. Failure to follow all shop safety regulations and procedures will result in a failure of this EC

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Enabling Check is to confirm the TPs covered in EO 002.02 and assess your layout skills by completing Template Layout Project IAW working drawings and specified tolerances;

2. You will be given:

   a. Mild steel stock pre-cut to 5" X 10", 1/8" thick; 5" X 10", 1/8" thick, one side square;

   b. Machinist tool box;

   c. Precision layout tools

   d. Shop fluids;

   e. Working drawing complete with EC checklist; and

   f. Supervision.

3. You will be denied assistance;

4. You will have 4 x 50 minute periods to complete EC 002.02;

5. Minimum pass mark is 75% IAW EC 002.02 marking guide;

6. You will be permitted a maximum of three restarts;

7. There is a 5% penalty for restarts;

8. Failure to achieve a passing mark will result in completion of Trainee Experiencing Difficulty Form; and

9. Observe and adhere to machine shop safety procedures, failure to do so will result in a failure of this EC.

## PO 002 Perform Machine Shop Fitting

## EO 002.03

1. Performance. Perform Hand Fitting Operations

2. Conditions.

   a. Given:

      (1) Operational shop equipment and hand tools;

      (2) Applicable stock;

      (3) References;

      (4) Supervision;

      (5) Student hand-outs; and

      (6) Precision layout tools to include:

         (a) surface plate

         (b) surface gauge,

         (c) hermaphrodite calipers,

         (d) dividers,

         (e) combination square,

         (f) vernier height gauge,

         (g) vernier protractor,

         (h) metal scriber;

      (7) Precision measuring tools to include:

         (a) Vernier measuring instruments;

         (b) micrometer measuring instruments;

            i. dial indicator;

            ii. thread gauges;

iii. thickness gauges;

iv. caliper - inside and outside;

v. telescopic gauges; and

vi. depth gauges.

(8) Machinist tool box.

b. Denied: Assistance;

c. Enviromental:

(1) Equipped machine/fitting shop.

3. Standard.

a. IAW the references, operating procedures and safety precautions, the student must explain and demonstrate the correct use of:

(1) Safety regulations and operating procedures;

(2) Types, purposes, descriptions and care of metalworking hand tools;

(3) Types, purposes, descriptions and care of metalworking power tools; and

(4) Student must complete PC project (Triangle and plate) IAW Ref. D3 (Naval Machinist Project Drawing Specifications).

4. Time.

a. 4 X 50 min periods for instruction

b. 44 x 50 min periods for PC

5. Method/Media.

a. Lectures;

b. Demonstrations;

c. Videos; and

d. Skills Practice.

6. References.

A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D3

7. Training Aids.

a. Layout tools specified in para. 2.a.;

b. Precision measuring tools specified in para. 2.a.

c. 2” x 2” x 1” inch mild steel stock (1 per student);

d. 2” x 2” x ½” inch mild steel stock (1 per student); and

e. Video presentation #386657 6306.

8. Test Details.

a. Practical PC: PC 002: Triangle and Plate marked by means of checklist

b. Minimum pass mark 75%

c. 44 x 50 min periods for PC

9. Remarks.

a. To assess the TPs of this PO, the assessment of this PO will consists of Triangle and Plate Project to be completed as per drawing specifications within time specified.

b. Accuracy as specified in the working drawing detailed in Ref D3.

10. Teaching Points.

a. Explain hand and power tool safety regulations and operating procedures to include;

(1) Hand tools to include:

(a) hammers;

(b) screw drivers;

(c) punches;

(d) pliers;

(e) wrenches;

(f) chisels; and

(g) machinist clamps.

(2) Hand cutting tools to include:

(a) hacksaws;

(b) twist drills;

(c) reamers; (straight, machine, adjustable and taper)

(d) center drills;

(e) counter bores;

(f) spot facers;

(g) files; ( degrees of coarseness and types

(h) taps and dies; die types(solid, split, adjustable, bevel, and thread chasing); tap types (taper, plug and bottoming); and

(i) broach.

(3) Power tools to include:

(a) electric drills;

(b) drilling machines;

(c) band saws; and

(d) bench grinders;

(4) Specialty items to include:

(a) wiggler gauge;

(b) lapping plates; and

(c) grinding and lapping compounds;

(5) Explain classes of fit/surface finishes to include:

(a) machine fit; and

(b) thread fit.

## Performance Check 002 – Triangle and Plate Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

### MATERIAL AND FACILITIES REQUIRED

1. Equipped machine shop
2. Machinist tool box;
3. Precision layout tools to include:
    a. surface plate
    b. surface gauge,
    c. hermaphrodite calipers,
    d. dividers,
    e. combination square,
    f. vernier height gauge,
    g. vernier protractor,
    h. metal scriber;
4. Precision measuring tools to include:
    a. Vernier measuring instruments;
    b. micrometer measuring instruments;
        (1) dial indicator;
        (2) thread gauges;
        (3) thickness gauges;
        (4) caliper - inside and outside;
        (5) telescopic gauges;
        (6) depth gauges; and
5. Metal Stock:
    a. 2” x 2” x 1” inch mild steel stock (1 per student); and

b. 2” x 2” x ½” inch mild steel stock (1 per student);

6. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees hand fitting skills by constructing Triangle and Plate as per working drawing and specified tolerances IAW ref D3;
2. Assistance is not permitted;
3. Supervision is mandatory;
4. Time allocated is 44 x 50 minute periods to complete PC 002;
5. Part II lists what each trainee will be given;
6. Trainees to be familiar with PC 002 checklist;
7. Pass mark is 75% of total marks IAW PC 002 marking guide;
8. There is a 5% penalty for restarts;
9. Each trainee will be permitted a maximum of three restarts;
10. There will be no re-writes PC 002, failure to achieve a passing mark for this PC will result in a TRB; and
11. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
12. PC 002 (Triangle and Plate Project) represents 5% of the final overall mark for Naval Machinist Course

**<u>PART II - INSTRUCTIONS GIVEN TO TRAINEE</u>**

1. This Performance Check is an evaluation of your hand fitting skills by constructing Triangle and Plate as per working drawing and specified tolerances.
2. You will be given:
   a. Machinist tool box;

b. Precision layout tools to include:

(1) surface plate

(2) surface gauge,

(3) hermaphrodite calipers,

(4) dividers,

(5) combination square,

(6) vernier height gauge,

(7) vernier protractor,

(8) metal scriber;

c. Precision measuring tools to include:

a. Vernier measuring instruments;

b. micrometer measuring instruments;

(a) dial indicator;

(b) thread gauges;

(c) thickness gauges;

(d) caliper - inside and outside;

(e) telescopic gauges;

(f) depth gauges; and

d. Metal Stock:

a. 2” x 2” x 1” inch mild steel stock (1 per student); and

b. 2” x 2” x ½” inch mild steel stock (1 per student);

e. Shop fluids; and

f. Supervision.

3. You will be denied assistance;

4. You will have 44 x 50 minute periods to complete this PC 002;

5. Minimum pass mark is 75% IAW PC 002 marking guide;

6. You will be permitted a maximum of three restarts for PC;

7. There is a 5% penalty for restarts;

8. There will be no re-writes for PC 002, failure to achieve a passing mark will result in a TRB; and

9. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.

## PO 003 Perform Advanced Grinding Techniques

## EO 003.01

1. Performance. Identify and Select Abrasive Wheels

2. Conditions.

   a. Given:

      (1) grinders;

      (2) machinist tool box;

      (3) eye protection;

      (4) silicon carbide abrasive wheels;

      (5) aluminum oxide abrasive wheels;

      (6) supervision; and

      (7) references.

   b. Denied: N/A

   c. Environmental:

   (1) Equipped machine shop

3. Standard.

   a. Describe the selection criteria IAW ref C5, for silicon carbide and aluminum oxide abrasive wheels.

   b. Observe and adhere to grinding safety procedures IAW ref C5,

4. Time.

   a. 3 X 50 minute period for instruction

5. Method/Media.

   a. Instruction

   b. Demonstration

c. Video; and

d. Homework assignment

6. References.

A1, C4, C5.

7. Training Aids.

a. Aluminum oxide and silicon carbide stone;

b. Grinder; and

c. Video presentation #38657 6307.

8. Test Details.

a. The TPs covered in EO 003.01 will be assessed on completion of PO 004 as PC 004 a written exam.

9. Remarks.

a. Trainees are required to read ref C5 section 16, before receiving instruction in PO 003.

b. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points.

a. Explain grinder safety and emphasize importance.

b. Describe grinding wheels:

(1) abrasive grain;

(2) bond types;

(3) grade;

(4) structure;

(5) common shapes and applications; and

(6) selection.

c) Explain grinding wheel marking system.

d) Describe handling and storage of abrasive wheels.

## PO 003 Perform Advanced Grinding Operations

## EO 003.02

1. Performance. Test and Mount Abrasive Wheels

2. Conditions.

   a. Given:

      (1) grinder

      (2) abrasive wheels

      (3) eye protection

      (4) machinist tool box

      (5) references

      (6) assistance, and

      (7) supervision

   b. Denied:

   c. Environmental:

3. Standard.

   a. IAW ref C5, test and mount abrasive wheels

4. Time.

   a. 4 X 50 minute periods for instruction

5. Method/Media.

   a. Instruction;

   b. Demonstration; and

   c. Skills Practice.

6. References.

   A1, C4, C5, D1.

7. Training Aids.

a. Grinder;

b. Abrasive wheels;

c. Machinist tool box; and

d. PPE.

8. Test Details.

a. The TPs covered in EO 003.02 will be assessed on completion of PO 004 as PC 004 (Written Examination).

9. Remarks.

a. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points.

a. Skills Practice: performed in conjunction with teaching points below

b. Explain grinding wheel speed ranges;

c. Explain grinding wheel torque limits;

d. Explain and demonstrate grinding wheel test methods (visual test and ring test);

e. Explain and demonstrate grinding wheel mounting procedures;

f. Skills Practice: Explain and demonstrate inspection techniques;

g. Skills Practice: Explain and demonstrate mounting procedures; and

h. Skills Practice: Explain and demonstrate dressing and truing of a grinding wheel.

**PO 003 Perform Advanced Grinding Operations.**

**EO 003.03**

1. Performance True and Dress Abrasive Wheels.

2. Conditions

   a. Given:

      (1) Grinders

      (2) Stone dressers

      (3) Machinist tool box

      (4) Eye protection

      (5) Silicone carbide abrasive wheels;

      (6) Aluminum oxide abrasive wheels;

      (7) Diamond dresser;

      (8) Supervision and

      (9) References

   b. Denied: N/a

   c. Enviromental:

      (1) Equipped machine shop

3. Standard:

   a. True and dress abrasive wheels IAW ref, C5

4. Time

   a. 1 x 50 min periods for instruction/Skills Practice

5. Method/media

   a. Instruction;

   b. Demonstration; and

c. Skills practice

6. References:

A1, C4, C5.

7. Training aids

a. Grinder;

b. Stone dresser;

c. Diamond dresser;

d. Abrasive wheels;

e. PPE; and

f. Machinist toolbox.

8. Test details

a. The TPs covered in EO 002.01 to be assessed on completion of PO 004 as PC 004 a written exam.

9. Remarks

a. Trainees to demonstrate truing and dressing procedures through practice.

b. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points

a. Skills Practice: Explain and demonstrate truing and dressing abrasive wheels.

## PO 003 Perform Advanced Grinding Operations.

### EO 003.04

1. Performance: Grind drill and tool bits

2. Conditions:

   a. Given:

      (1) Grinders

      (2) Stone dressers

      (3) Machinist tool box

      (4) Eye protection

      (5) Silicone carbide abrasive wheels;

      (6) Aluminum oxide abrasive wheels;

      (7) Diamond dresser;

      (8) Tool bits;

      (9) Drill bits;

      (10) $59^0$ drill point gauge;

      (11) $60^0$ centre gauge

      (12) Supervision and

      (13) References

   b. Denied: N/A

   c. Enviromental:

      (1) Equipped machine shop

3. Standard:

   a. IAW ref, C5, grind a drill bit to 59 deg. (118deg included)

   b. IAW ref, C5, grind a threading tool to 60 deg.

4. Time

   a. 2 x 50 min periods for instruction.

   b. 1 X 50 min periods for skills Practice

5. Method/media

   a. Instruction;

   b. Demonstration; and

   c. Skills Practice.

6. References:

   A1, C4, C5.

7. Training aids

   a. Grinders

   b. Stone dressers

   c. Machinist tool box

   d. Eye protection

   e. Silicone carbide abrasive wheels;

   f. Aluminum oxide abrasive wheels;

   g. Diamond dresser;

   h. Tool bits;

   i. Drill bits;

   j. $59^0$ drill point gauge;

   k. $60^0$ centre gauge

   l. Videos: #38658 6318, #38658 6319 and #38657 6308

8. Test details

   a. This EO will be assessed on completion of PO 004 as PC 004 (Written Examination).

9. Remarks

a. Skills Practice: Trainees shall grind a drill bit 59 deg (with 118 deg included angle) and grind a threading tool to 60 deg.

b. Instructor to check trainee drill and threading tool bits with a 59 deg drill gauge and a 60 deg centre gauge respectively.

c. Trainees will use an HSS drill bit and blank tool stock to perform tool bit and threading tool bit grinding respectively.

d. Trainees will be required to read ref C5 as homework assignment, before instruction.

c. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points

a. Skills Practice: performed in conjunction with teaching points below.

b. Explain and demonstrate how to examine a drill bit to include:

(1) Length of cutting tip;

(2) Angle of cutting tips; and

(3) Free from nicks and wear.

c. Explain the causes of drill bit failure.

d. Drill and drill grinding jig procedure.

e. Drill point angles for various materials.

f. Visual examination of tool bit to include:

(1) Physical condition;

(2) Appearance;

(3) Correct relief angles/side rake.

g. Causes of tool bit failure to include:

(1) Improper relief and rake angles;

(2) Improper selection of tool bit material;

(3) Too high feed and speed;

(4) No cooling supplied.

h. Skills Practice: Grinding tool bits employing the use of grinding gauges.

## PO 004 Perform Precision Drill Press Operations

## EO 004.01

1. Performance. Prepare Work Piece For Drilling.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Supervision;

      (4) Metal stock;

      (5) Tool holding devices to include:

         (a) jacob chuck,

         (b) sleeve and socket;

      (6) Work holding devices to include:

         (a) contour vice,

         (b) V block,

         (c) angle vice,

         (d) angle plate,

         (e) finger clamps,

         (f) U - clamps,

         (g) C - clamps,

         (h) straight clamps,

         (i) step blocks,

         (j) parallel bars,

         (k) drill jigs, and

(l) rotary table;

(7) Alignment tools to include:

(a) centre finder,

(b) centre drill,

(c) drill bit,

(d) square, and

(e) dial indicator; and

(8) drill press.

b. Denied: n/a

c. Environmental:

(1) Equipped Machine Shop

3. Standard.

a. Observe and adhere to all safety rules and regulations IAW ref A1, C5.

b. IAW references, demonstrate set-up procedures of tools and equipment listed in para. 2.a.

c. Calculate the spindle speed using the cutting speed from ref C4, and the formula SS = (CS x 4) / D

4. Time.

a. 4 X 50 minute period for instruction.

5. Method/Media.

a. Lecture;

b. Demonstration

c. Video; and

d. Skills Practice.

6. References.

A1, C4, C5,

7. Training Aids.

a. See para 2.a. (2) to (6).

b. Video #38657 6310

8. Test Details.

a. The TPs covered in EO 004.01 will be assessed on completion of PO 004 as PC 004 (Written Examination).

9. Remarks.

a. All tools to be displayed and provisions made to allow the trainee to handle them.

b. The trainee will demonstrate knowledge of the material by calculating the feed rate and spindle speed of a 1/2" standard drill bit working with mild steel stock.

c. The use of calculators permitted.

d. PC 004 is an evaluation of TPs covered in PO 001, PO 003 and PO 004

10. Teaching Points.

a. Explain the significance of feeds and speeds.

b. Explain the factors that affect feeds and speeds.

c. Skills Practice: Calculate cutting speeds and feed rates using formulas in refs C5.

d. Explain the purpose of cutting fluids.

e. Explain the basic types of cutting fluids and uses.

f. Emphasize the importance of rigid and secure holding arrangement for the work piece.

g. Demonstrate and explain the operation and application of the holding

devices;

h. Demonstrate alignment procedures employing the following devices:

(1) centre finder;

(2) centre drill;

(3) machinists' square;

(4) drill bit; and

(5) dial indicator.

i. Demonstrate the operation and application of the following tool holding devices:

(1) jacob chuck; and

(2) drill sleeve.

## PO 004 Perform Precision Drill Press Operations

## EO 004.02

1. Performance. Use Specialty Cutters.

2. Conditions.

   a. Given:

      (1) Drill press;

      (2) PPE;

      (3) References;

      (4) Specialty cutters to include:

         (a) counter bore;

         (b) counter sink;

         (c) boring bar,

         (d) saw type hole cutter;

         (e) fly cutter;

         (f) spot facer;

         (g) reamer;

         (h) taps; and

         (i) step drills.

      (5) Assistance;

      (6) Supervision; and

      (7) Metal stock

   b. Denied: n/a

   c. Environmental:

      (1) Equipped machine shop.

3. Standard.

a. IAW refs :

(1) Visually identify and describe operations of specialty cutters;

(2) Describe the application of specialty cutters in para 2.a; and

(3) Observe and adhere to all safety rules and regulations.

4. Time.

a. 4 X 50 minute periods for instruction

b. 2 X 50 minute periods for PC

c. 2 X 50 minute periods for Pre/Post review

5. Method/Media.

a. Instruction;

b. Video;

c. Demonstration;

d. Skills Practice; and

e. Homework assignment (reading).

6. References.

A1, C4, C5.

7. Training Aids.

a. See para 2.a.

b. Video # 38657 6310.

8. Test Details.

a. The TPs covered in EO 004.02 to be confirmed as PC 004 (Written Examination).

b. Minimum pass mark 75%

c. 2 x 50 minute periods for PC

9. Remarks.

a. Instructor will carry out instruction in the machine shop and encourage the trainee to handle all training aids.

b. Trainee will read ref C5, unit 22 and unit 23 prior to commencement of EO 004.02.

c. Instructor will have the specialty cutters listed in para 2d on display and will conduct an oral brief on the type of cutters displayed and functions of each.

d. PC 004 will include all TPs covered in PO 001, PO 002 and PO 004.

10. Teaching Points.

a. Describe types, application and operation of specialty cutters to include:

(1) step drill;

(2) reamer (power);

(3) spot facers;

(4) counter sink;

(5) boring bar;

(6) counter bore;

(7) saw type hole cutter;

(8) fly cutter; and

(9) taps.

b. Demonstrate the set-up and tool holding methods for each cutter.

## Performance Check 004 – PO 004 Drill Press Operations; PO 003 Advance Grinding Techniques; PO 001 Machine Shop Management

### PART I - ADMINISTRATIVE INSTRUCTIONS

### MATERIAL AND FACILITIES REQUIRED

1. Classroom with seating for 6 trainees.

### ARRANGEMENTS AND PROCEDURES

1. This PC will assess the TPs covered in PO 001 - Machine Shop Management, PO 003 - Advance Grinding Techniques and PO 004 - Drill Press Operations.
2. This PC is a written examination.
3. The pass mark for this PC is 75%.
4. Assistance and References during this PC are not permitted.
5. Trainees will have 2 x 50 minute periods to complete PC.
6. Failure to achieve a passing mark will result in a failure for this PC, trainees will be afforded oppurtunity to a supplementary exam, failure of the supplementary exam will result in a TRB.
7. PC 004 will represent 5% of the final overall mark for Naval Machinist Course.

### PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. The PC will cover the concepts of Machine Shop Management, Advance Grinding Techniques and Drill Press Operations.
2. This is a written PC.
3. Pass mark for this PC is 75%.
4. Assistance and references during this PC are not permitted.
5. You have 2 x 50 minute periods to complete this PC.
6. Failure to achieve a passing mark for this PC will result in a TRB.

## PART III - PERFORMANCE CHECK NOTES

1. PC 004 will assess the TPs covered in PO 001 Manage Machine Shop; PO 003 Advanced Grinding Techniques and PO 004 Advanced Drill Press Operations

## PO 005 Perform Advanced Machine Lathe Operations

### EO 005.01

1. Performance. Calculate Spindle Speeds And Feed Rates.

2. Conditions.

   a. Given:

      (1) references;

      (2) assistance;

      (3) supervision; and

      (4) Lathe with digital readout.

   b. Denied:

   c. Environmental:

      (1) Equipped Machine Shop

3. Standard.

   a. IAW ref C4, calculate the spindle speeds and feed rates:

      (1) spindle speed using the formula: SS = (4 x CS) / D

      (2) rate of feed (V) using the formula: V = (D x SS) / 4

   b. IAW DRO manual, demonstrate ability to use digital readout.

4. Time.

   a. 2 x 50 minute periods for instruction.

5. Method/Media.

   a. Lecture;

   b. Demonstration

   c. Video

   d. Skills Practice

e. Homework assignment (reading)

6. References.

A1, C3, C4, D2

7. Aids.

a. Video #38658 6317;

b. Lathe with digital readout; and

c. metal stock.

8. Test Details.

a. Class assignment (see para 10).

b. The TPs covered in EO 005.01 will be confirmed on completion of EO 005.04 by means of practical EC 005.04 Stud and Nuts Project.

9. Remarks.

a. The trainee will demonstrate knowledge of the material by calculating the feed rate and spindle speed of an 8" diameter, piece of brass stock.

b. Instruction will take place in the classroom.

c. Calculators are permitted.

d. Trainee is required to read ref C5, unit 27 and ref D2 before commencement of EO 005.01.

e. Students shall not be permitted the use of digital readout devices until the completion of EC 005.05 (Valve Spindle).

10. Teaching Points.

a. Explain the importance of feeds and speeds on the lathe.

b. Explain the factors that affect the feeds and speeds.

c. Skills Practice: calculate cutting speeds and feed rates using the formula stated in the standard, para 3

d. Skills Practice: operate the digital readout IAW ref D2

## PO 005 Perform Advanced Machine Shop Operations

## EO 005.02

1. Performance. Carry Out Alignment and Setup Procedure.

2. Conditions.

   a. Given:

      (1) references;

      (2) lathe;

      (3) mandrel;

      (4) dial indicator;

      (5) metal stock;

      (6) machinist tool box;

      (7) assistance; and

      (8) supervision.

   b. Denied:

   c. Environmental:

      (1) equipped machine shop

3. Standard.

   a. IAW ref C5, align the tailstock within 0.001".

   b. IAW ref C1, observe and adhere to all lathe safety operations.

4. Time.

   a. 4 X 50 minute periods for demonstration and skills practice.

5. Method/Media.

   a. Demonstration

   b. Video

c. Skills Practice

d. Homework assignment: (reading)

6. References.

A1, C1, C5,

7. Training Aids.

a. See para 2.a; and

b. Videos #38658 6323.

8. Test Details.

a. The TPs covered in EO 005.02 to be confirmed on completion of EO 005.04 as practical EC 005.04 Stud and Nuts Project.

9. Remarks.

a. This EO to be conducted in the machine shop with unlimited access to the engine lathes.

b. All necessary tools and equipment will be set-up and displayed prior to commencing this EO.

c. The instructor shall ensure that student participation and “hands on" is given at the earliest opportunity.

d. The trainee is required to read ref C1, C5 prior to the commencement of EO 005.02.

e. This EO will not be tested; the trainee is to demonstrate to instructor the ability to align tailstock to 0.001 inches.

10. Teaching Points.

a. Explain and demonstrate the following lathe controls for review:

(1) master power isolation;

(2) quick change gear-box controls;

(3) carriage controls;

(4) cross-feed controls;

(5) auto feed controls on apron;

(6) threading controls;

(7) cutting oil reservoir and controls; and

(8) safety devices.

b. Explain the requirement to ensure/check lathe tailstock alignment;

c. Explain and demonstrate the four techniques used ensure tailstock alignment IAW ref C1 and C5;

d. Demonstrate and explain the simplest and most accurate method to align the tailstock with the mandrel and dial indicator;

e. Ensure that emphasis is placed on time and accuracy considerations when choosing the method of tailstock alignment; and

f. Explain the safety precautions associated with engine lathe to include:

(1) use of safety glasses;

(2) running of lathe.

## PO 005 Perform Advanced Machine Lathe Operations

## EO 005.03

1. Performance. Mount Lathe Attachments

2. Conditions.

   a. Given:

      (1) Lathe;

      (2) tool posts and holders;

      (3) lathe dogs;

      (4) dog driving plate;

      (5) face plate;

      (6) three and four jaw chucks;

      (7) Jacob chuck;

      (8) collets and chuck;

      (9) follower rest;

      (10) steady rest;

      (11) taper attachment;

      (12) milling attachment:

         (a) rotary table;

         (b) tool post grinder;

      (13) machinist tool box;

      (14) references,

      (15) assistance, and

      (16) supervision.

   b. Denied: n/a

c. Environmental:

(1) Equipped machine shop

3. Standard.

a. IAW ref C5, unit 28, 29, 30, 32 and 33, demonstrate the use of the various lathe attachments with strict emphasis on safety.

4. Time.

a. 3 X 50 minute periods for instruction

b. 2 x 50 minute periods for demonstration.

5. Method/Media.

a. Instruction

b. Demonstration

c. Video; and

d. Homework assignment (reading).

6. References.

A1, C1, C5

7. Training Aids.

a. See para 2.a.

b. Video presentation, #38659 6340, #38659 6341, #38659 6342, #38658 6316, #38658 6324 and #38658 6326.

8. Test Details.

a. The TPs covered in EO 005.03 will be confirmed on completion of EO 005.04 by means of practical EC 005.04 Stud and Nuts Project.

9. Remarks.

a. All lathe attachments are to be displayed to the students and set up prior to commencing instruction.

b. The trainee must decide which lathe accessories they will employ with consideration for accuracy, safety, time and simplicity.

c. Trainee will read ref C5, unit 32 and 33 prior to commencement of EO 005.03.

10. Teaching Points.

a. Explain and demonstrate the common attachments that are utilized on the lathe;

b. Explain the applications and limitations which must be considered when choosing an attachment;

c. Explain that while two attachments may do the same job, the machinist must consider accuracy, time, simplicity, rigidity and safety when making a choice;

d. Lathe attachments to be discussed shall include:

(1) tool posts and holders;

(2) lathe dogs and plate;

(3) milling attachment;

(4) rotary table;

(5) face plate;

(6) three and four jaw chuck;

(7) collet chuck;

(8) jacob chuck;

(9) follower rest;

(10) steady rest;

(11) taper attachment; and

(12) tool post grinder.

e. Demonstrate the mounting procedure for each individual lathe attachment.

## PO 005 Perform Advanced Machine Lathe Operations

## EO 005.04

1. Performance. Perform Lathe Operations.

2. Conditions.

   a. Given:

      (1) lathe

      (2) machinist tool box

      (3) metal stock

      (4) references

      (5) assistance, and

      (6) supervision

   b. Denied: n/a

   c. Environmental:

      (1) Equipped machine shop.

3. Standard.

   a. Observe and adhere to shop safety procedures IAW reference, and

   b. Machine the lathe projects to specified tolerances IAW ref D3 (Specification Drawings).

4. Time.

   a. 43 x 50 Minute periods for Skills Practice

   b. 41 x 50 Minute periods for Practical EC

5. Method/Media.

   a. Videos

   b. Skills Practice

6. References.

A1, C1, C4, C5, D3

7. Training Aids.

a. See para 2.a

b. Videos 38658 6328, 38659 6334, 38659 6335, 38659 6336, 38659 6337, and 38659 6339.

8. Test Details.

a. Practical EC 005.04: Stud and Nuts Project

b. Minimum pass mark is 75% IAW checklist.

c. 41 x 50 Minute periods for Practical EC

9. Remarks.

a. Instructor will assist and periodically check trainee progress,

b. Trainee will machine, for practice, the Turning and Threading Lathe Projects IAW ref D3. Upon completion of the instructor's final check, trainee will part three (3) washers from the bored end of the Turning Project. Instructor will check the final product.

c. Practical EC 005.04: consists of a practical project “Stud and Nuts Project” IAW ref D3 administered to each trainee by instructor to confirm TPs covered in EO 005.01 thru to EO 005.04.

d. Students shall not be permitted the use of digital readout devices until the completion of EC 005.05 (Valve Spindle).

10. Teaching Points.

N/A

## Enabling Check 005.04 – Stud and Nuts Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop
2. Machinist tool box;
3. Precision layout tools
4. Precision measuring tools
5. Metal Stock:
   a. 1 ¼" Round, Hot Rolled Mild Steel, 4 ½" Long (1 per student).
   b. 1 ¾" Mild Steel Hex stock, 2" Long (1 per student).
6. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Enabling Check is to confirm TPs covered in EO 005.04 by machining Stud and Nuts IAW ref D3 (working drawing and specified tolerances);
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 41 x 50 minute periods to complete EC 005.04.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with EC checklist.
7. Pass mark is 75% of total marks IAW EC 005.04 marking guide.
8. There is a 5% penalty for restarts.
9. Each trainee will be permitted a maximum of three restarts.
10. Failure to achieve a passing mark of 75% for this EC will result in completion of Trainee Experiencing Difficulty form.

11. Failure to follow all shop safety regulations and procedures will result in a failure of this EC.

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Enabling Check is to confirm the TPs covered in EO 005.04 by machining Valve Spindle Project IAW working drawings and specified tolerances.
2. You will be given:
   a. Metal Stock:
      (1) 1 ¼" Round, Hot Rolled Mild Steel, 4 ½" Long (1 per student)
      (2) 1 ¾" Mild Steel Hex stock, 2" Long (1 per student).
   b. Machinist tool box;
   c. Precision layout tools
   d. Shop fluids;
   e. Working drawing complete with EC checklist; and
   f. Supervision.
3. You will be denied assistance
4. You will have 41 x 50 minute periods to complete EC;
5. Minimum pass mark is 75% IAW EC 005.04 marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. Failure to achieve a passing mark will result in completion of Trainee Experiencing Difficulty Form
9. Observe and adhere to machine shop safety procedures, failure to do so will result in a failure of this EC.

**PART III - ENABLING CHECK NOTES**

1. Dimension 13 to be measured by one wire method (use 0.4872P). For the bottom

end tolerance the wire must not be loose between the micrometer anvil and the thread form.

2. Both nuts are to be cut to fit the stud;
3. Lengths of raw stock given are to give trainees for developing planning work skills and experience working with short piece work;
4. Trainees to use High Speed Cutting Tools only;
5. Digital measuring tools and digital read-out apparatus not permitted;
6. Dimensions 7 and 14 to be finished by hand.
7. Project must be complete and assembled for passing mark.

## PO 005 Perform Advanced Machine Lathe Operations

## EO 005.05

1. Performance. Perform Precision Lathe Operation

2. Conditions.

   a. Given:

      (1) lathe

      (2) machinist's tool box,

      (3) metal stock material,

      (4) references,

      (5) assistance, and

      (6) supervision.

   b. Denied: n/a

   c. Environmental:

   a. Equipped machine shop

3. Standard.

   a. Observe and adhere to shop safety procedures

   b. Machine the Mini Bomb Project to specified tolerances IAW ref D3

4. Time.

   a. 2 x 50 min periods for instruction/demonstration;

   b. 34 x 50 minute periods for EC; and

   c. 39 x 50 minute periods for PC.

5. Method/Media.

   a. Lecture;

   b. Demonstration; and

c. Skills Practice.

6. References.

A1, C1, C3, C4, C5, D3

7. Training Aids.

a. See para 2.a

8. Test Details.

a. Practical EC: Machine Valve Spindle Project

b. Practical PC: Machine Mini Bomb Project.

c. Pass mark 75% as per PC/EC checklists.

d. 34 x 50 minute periods for EC

e. 39 x 50 minute periods for PC

9. Remarks.

a. Instructor will assist and periodically check trainee progress,

b. Trainee required to review ref C1, Ch 11 and, C5 sect 11, prior to commencement of EO 005.05.

c. Practical EC: Trainee will machine the Valve Spindle Project IAW ref D3. Instructor will mark the final product IAW checklist. It is a practical project to confirm TP covered in EO 005.05 to be completed prior to commencing PC.

d. Students are permitted to use the Digital Readout Devices used at instructors discretion on the completion of the Valve Spindle.

e. Practical PC: The PC for this PO will be to Machine the Mini Bomb Project to specified tolerances IAW ref D3, to assess TPs covered in PO 005. The instructor must emphasize that safety is a critical priority while performing this PC.

10. Teaching Points.

a. Demonstrate and explain:

(1) calculating the dimensions of various keyways and the operation of broaching / milling a keyway on the lathe.

(2) turning work between centres;

(3) knurling operations on the engine lathe;

(4) internal and external threading operations on the engine lathe;

(5) various taper turning operations;

(6) multi-start threading operations on the engine lathe;

(7) all unique set-up operations and considerations which apply to each different machining operations; and

(8) all safety considerations when employing the use of advanced lathe tools and accessories beginning with the initial set-up, to the completion of the particular machining operation.

## Enabling Check 005.05 - Valve Spindle Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop
2. Machinist tool box;
3. Precision layout tools to include:
4. Precision measuring tools to include:
5. Metal Stock:
   a. 2” Round, Hot Rolled Mild Steel, 19” Long (1 per student).
6. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Enabling check is to confirm TPs covered in EO 005.05 by machining a Valve Spindle IAW ref D3 (working drawing and specified tolerances);
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 34 x 50 minute periods to complete EC.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with EC checklist.
7. Pass mark is 75% of total marks in marking guide.
8. There is a 5% penalty for restarts.
9. Each trainee will be permitted a maximum of three restarts.
10. Failure to achieve a passing mark of 75% for this EC will result in completion of Trainee Experiencing Difficulty form.
11. Failure to follow all shop safety regulations and procedures will result in a failure of

this EC.

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Enabling Check will confirm the TPs covered in EO 005.05 and assess your lathe operating abilities by machining Valve Spindle Project IAW working drawings and specified tolerances.
2. You will be given:
   a. Equipped machine shop: Lathe equipped with attachments;
   b. Metal Stock: 2” Round, Hot Rolled Mild Steel, 19” long;
   c. Machinist tool box;
   d. Precision layout tools
   e. Shop fluids;
   f. Working drawing complete with EC checklist; and
   g. Supervision.
3. You will be denied assistance
4. You will have 4 x 50 minute periods to complete EC;
5. Minimum pass mark is 75% IAW marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. Failure to achieve a passing mark will result in completion of Trainee Experiencing Difficulty Form
9. Observe and adhere to machine shop safety procedures, failure to do so will result in a failure of this EC.

**PART III - ENABLING CHECK NOTES**

1. Dimension 6 major diameter to be machined to limits of dimension 2;
2. Dimension 6 pitch diameter to be measured by one wire method as per ref C5, pg

266 (Technology of Machine Tools);

3. Use #40 wire (Drill Bit Shank) for best wire size);
4. For dimension 8 use 3 wire method or thread micrometer;
5. Dimension 7 to be filed square;
6. High Speed steel (HSS) cutting tools to be used only;
7. The use of Digital measuring tools and digital read-out not permitted;
8. External thread forms shall be cut single-point on the lathe: Die nuts or other aids not permitted.

## Performance Check 005 – Mini Bomb Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop;
2. Machinist tool box;
3. Precision layout tools;
4. Precision measuring tools;
5. Metal Stock:
   a. 1” Hex stock, mild steel, 7/8” long (1 per student); and
   b. 1 ½ round, Hot Rolled Mild Steel, 6” Long (1 per student); and
   c. 2” Round, Hot Rolled Mild Steel, 1 3/8” Long (1 per student).
6. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees lathe operating skills by machining a bomb as per working drawing and specified tolerances IAW ref D3;
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 39 x 50 minute periods to complete PC.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with PC checklist.
7. Pass mark is 75% of total marks in marking guide.
8. Each trainee will be permitted a maximum of three restarts.
9. There is a 5% penalty for restarts;
10. There will be no re-writes of this PC for this PO; failure to achieve a passing mark for

this PC will result in a TRB.

11. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
12. PC 005 Mini Bomb represents 10% of the final overall mark for Naval Machinist Course

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Performance Check is an evaluation of your lathe operating abilities.
2. You will be given:
    a. Machinist tool box;
    b. Precision layout tools to include:
    c. Precision measuring tools to include:
    d. Metal Stock, pre-cut to rough lengths;
    e. Shop fluids; and
    f. Supervision.
3. You will be denied assistance
4. You will have 44 x 50 minute periods to complete this PC;
5. Minimum pass mark is 75% IAW marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. There will be no re-writes for this PC, Failure to achieve a passing mark will result in a TRB; and
9. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
10. PC 005 Mini Bomb represents 10% of the final overall mark for Naval Machinist Course

**PART III - PERFORMANCE CHECK NOTES**

1. The instructor must emphasize that safety is a critical priority while performing this PC.
2. Raw steel is deliberately supplied as “short cuts”;
3. Straight sleeve must fit two ways: stud end and towards taper sleeve;
4. Project must be complete and assembled for pass;
5. Digital measuring tools are not permitted
6. Digital read-out displays used at instructors discretion.

## PO 006 Perform Advanced Milling Machine Operations

## EO 006.01

1. Performance. Demonstrate Basic Operating Procedures Of The Milling Machine And Attachments.

2. Conditions.

   a. Given:

      (1) Milling machine,

      (2) Machinist's tool box,

      (3) Collets, arbors, and adaptors,

      (4) Various vices,

      (5) Rotary table,

      (6) Index head,

      (7) Vertical milling attachment,

      (8) Clamps and fasteners,

      (9) References,

      (10) Assistance,

      (11) Supervision.

   b. Denied: N/A.

   c. Environmental:

      (1) Equipped machine shop.

3. Standard.

   a. Observe and adhere to milling machine safety procedures IAW ref C5,

   b. Demonstrate milling machine operating procedures IAW ref C5,

   c. Install milling machine attachments IAW ref C5,

4. Time.

a. 5 x 50 minute periods for instruction

5. Method/Media.

a. Video;

b. Instruction;

c. Demonstration; and

d. Skills Practice

6. References.

A1, A2, A3, C1, C2, C3, C4, C5

7. Training Aids.

a. Refer to para 2a and Annex H, para 3b

b. Videos 38660 6343, 38660 6348 and 38660 6349.

8. Test Details.

a. The TPs covered in EO 006.01 will be confirmed by means of practical EC 006.03 administered on completion of EO 006.03

9. Remarks.

a. The instructor will first demonstrate and explain the operation of the milling machine controls. The student will then be given 30 minutes to familiarize himself with the operations. At the end of this time, each student will display competent, control operation to the instructor. This must be done before the student is allowed to proceed with milling operations.

b. Each candidate will also identify and state the use of milling machine attachments.

c. Trainee required to read (home reading assignment), Milling Machine

Operating Instruction Manual prior to commencing EO 006.01

10. Teaching Points.

a. Explain and demonstrate each fixed component of a milling machine. (Ref. C5, unit 36 pp 252-259).

b. Explain and demonstrate operating controls for the milling machine to include:

(1) power isolation switch,

(2) on-off switch,

(3) cutter rotation switch,

(4) table and crossfeed hand wheels,

(5) quick change feed and speed controls,

(6) cutting fluid reservoir and controls,

(7) power traverse controls,

(8) table locks,

(9) feed stop,

(10) rapid power traverse,

(11) power transmission compartment;.

c. Explain and demonstrate the application and mounting techniques for the following:

(1) collars, adaptors, arbors,

(2) work holding devices,

(3) rotary table,

(4) indexing head,

(5) vertical milling attachment ,

(6) clamps and fasteners.

d. Identify and explain all safety hazards for mounting procedures set-up and operation of the milling machine.

## PO 006 Perform Advanced Milling Machine Operations

## EO 006.02

1. Performance. Mount Milling Cutters.

2. Conditions.

   a. Given:

      (1) milling machine;

      (2) milling cutters;

      (3) machinist's tool box;

      (4) references;

      (5) assistance; and

      (6) supervision.

   b. Denied: n/a

   c. Environmental:

   (1) Equipped machine shop

3. Standard.

   a. Classify milling cutters with respect to types and applications IAW ref C5;

   b. Mount the following, obeying and adhering to safety regulations for handling and mounting cutters, ref C5:

      (1) vertical cutter; and

      (2) horizontal cutter.

   c. Describe milling cutters, IAW para 10, to include:

      (1) care of, including storage and handling,

      (2) mounting/dismounting procedures, and

      (3) cutting fluids.

4. Time.

a. 4 X 50 minute periods for instruction

5. Method/Media.

a. Instruction;

b. Video;

c. Demonstration; and

d. Home work assignment (reading)

6. References.

C1, C3, C5.

7. Training Aids:

a. Plain mill cutter;

b. Side mill cutter;

c. Angular cutter;

d. Formed cutter;

e. Gear cutter;

f. Shell end cutter;

g. Arbors;

h. Vertical milling attachment;

i. end mill cutter;

j. machinist tool box;

k. Storage facilities for cutters;

l. Videos 38660 6344, 38660 6347 and 38660 6350;

m. slit saw cutter;

n. slab mill cutter; and

o. non-HSS cutters.

8. Test Details.

a. The TPs covered in EO 006.02 will be confirmed by means of practical EC 006.03 administered on completion of EO 006.03.

9. Remarks.

a. The instructor shall stress all safety consideration, regulations throughout this EO.

b. The instructor will set up a display of the various milling cutters prior to commencing instruction.

c. All instruction will take place in the machine shop with the exception of calculating feeds and speeds.

d. The trainee must demonstrate to the instructor the following:

(1) the method of lifting the attachment;

(2) how to prepare both machine and attachment for installation; and

(3) Installation of attachments will be done in groups of two (2) trainees.

e. Emphasis must be stressed during the installation of work holding devices and mill cutters on safety, accuracy, finish and time.

f. Trainee required to read ref C5, unit 36 prior to commencement of EO 006.02.

10. Teaching Points.

a. Explain the reasons why machine speeds and material feed rates are critical and are calculated;

b. Demonstrate and explain the method of calculating the feed rate and cutter speed, using the following formulas:

Feed = N X CPT X Rev/min

Rev/min = r X CS/Dia

c. Identify and describe the various milling cutters available which include:

(1) plain milling cutters;

(2) side milling cutters;

(3) angular cutters;

(4) formed cutters;

(5) gear cutters;

(6) shell end mill cutters;

(7) Slot Cutters;

(8) slit saw;

(9) slab mill cutter; and,

(10) non-HSS cutters.

d. Explain the applications and limitations of each cutter considering accuracy, safety, finish and time;

e. Explain the care and storage of milling cutters;

f. Explain the three (3) general categories and applications of cutting fluids which include:

(1) cutting oils;

(2) emulsifiable or soluble oils; and

(3) chemical cutting fluids.

g. Explain and demonstrate application, handling and mounting/dismounting procedures for each milling cutter and arbors with emphasis placed on safety.

## PO 006 Perform Advanced Milling Machine Operations

## EO 006.03

1. Performance. Perform Basic Milling Operations.

2. Conditions.

   a. Given:

      (1) milling machine,

      (2) machinist's tool box,

      (3) stock,,

      (4) hold down devices,

      (5) references,

      (6) project drawing,

      (7) various milling cutters,

      (8) assistance, and

      (9) supervision.

   b. Denied: n/a

   c. Environmental:

      (1) equipped machine shop

3. Standard.

   a. Select, install and align applicable holding device IAW ref C5,

   b. Select and install milling cutter IAW ref C3,

   c. Following procedures IAW ref C5, set-up metal stock for milling.

   d. Mill a 2" square metal cube to tolerances as per supplied working drawing IAW D3.

   e. Observe and obey all milling machine safety procedures IAW ref C5.

4. Time.

a. 25 x 50 minute periods instruction/skills practice, and

b. 8 X 50 minute periods for EC (basic milling project block).

5. Method/Media.

a. Instruction;

b. Demonstration; and

c. Skills practice.

6. References.

C1, C3, C5, D3.

7. Training Aids.

a. Vice

b. Milling cutters

8. Test Details.

a. Practical EC 006.03: Basic Milling project to be completed as per drawing and specifications IAW Ref. D3

9. Remarks.

a. Trainee required to read ref. C5 prior to instruction

b. The instructor will ensure that a display of all training aids is set up prior to instruction.

c. Emphasis must be stressed during the installation of work holding devices and mill cutters on safety, accuracy, finish and time.

10. Teaching Points.

a. Skills Practice performed in concert with teaching points below;

b. Demonstrate and explain the various material and cutter set-ups on the milling machine:

(1) aligning a vice;

(2) setting the cutter to the work surface; and

(3) machine set-ups and operations.

c. Explain conventional and climb milling.

## Enabling Check 006.03 - Basic Milling Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop to include milling machine and attachment;
2. Machinist tool box;
3. Hold down devices
4. various milling cutters,
5. Precision layout tools;
6. Precision measuring tools;
7. Metal Stock:
   a. 2 ¼" Square, Mild Steel, 2 1/8" Long (1 per student).
8. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Enabling check is to confirm TPs covered in EO 005.05 by completing a Basic Milling Project IAW ref D3 (working drawing and specified tolerances);
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 8 x 50 minute periods to complete EC 006.03.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with EC 006.03 checklist.
7. Pass mark is 75% of total marks IAW EC 006.03 marking guide.
8. Each trainee will be permitted a maximum of three restarts.
9. There is a 5% penalty for restarts.
10. Failure to achieve a passing mark of 75% for this EC will result in completion of

Trainee Experiencing Difficulty form.

11. Failure to follow all shop safety regulations and procedures will result in a failure of this EC.

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Enabling Check is to confirm the TPs covered in EO 006.03 and assess your lathe operating abilities by machining Valve Spindle Project IAW working drawings and specified tolerances.
2. You will be given:
    a. Milling Machine
    b. Hold down devices;
    c. Various cutters;
    d. Metal Stock: 2” Round, Hot Rolled Mild Steel, 19” long;
    e. Machinist tool box;
    f. Precision layout tools
    g. Shop fluids;
    h. Working drawing complete with EC checklist; and
    i. Supervision.
3. You will be denied assistance
4. You will have 4 x 50 minute periods to complete EC 006.03;
5. Minimum pass mark is 75% IAW EC 006.03 marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. Failure to achieve a passing mark will result in completion of Trainee Experiencing Difficulty Form
9. Observe and adhere to machine shop safety procedures, failure to do so will result in

**PART III - ENABLING CHECK NOTES**

1. Horizontal milling only;
2. High speed steel cutters only
3. Digital measuring tools and digital displays to be used at instructors discretion;
4. Individual trainee project: trainees are not to share milling machines or set-ups with other trainees.

## PO 006 Perform Advanced Milling Machine Operations

## EO 006.04

1. Performance. ` Machine Spur Gears
2. Conditions.
   a. Given:
      (1) milling machine;
      (2) machinist's tool box;
      (3) Metal stock;metal Stock;
      (4) indexing head;
      (5) gear tooth calipers;
      (6) micrometers;
      (7) gearing display;
      (8) scientific calculator;
      (9) references;
      (10) assistance; and
      (11) supervision.
   b. Denied: n/a

c. Environmental:

(1) Equiped machine shop

3. Standard.

a. Describe gear cutting theory, types of gears and terminology.

b. Machine a spur gear to specified tolerances IAW ref D3.

c. Adhere to all safety and operating procedures, ref C5,

4. Time.

a. 6 x 50 Minute Periods for Instruction

b. 10 x 50 minute Periods for Skills Practice

5. Method/Media.

a. Instruction;

b. Demonstration; and

c. Practice

6. References.

A1, C1, C4, C5, D3

7. Training Aids.

a. See para 2a .

8. Test Details.

a. The TPs covered in EO 006.04 will be assessed as Practical PC 007 (Gear Pump Project).

9. Remarks.

a. The student shall read ref, C5 prior to instruction.

b. The practice for this EO will consist of machining an 8 DP, 2.25 inch spur gear in aluminum stock as per drawing and specifications IAW D3

c. The instructor shall check the tolerances of each trainee's spur gear.

d. The instructor shall stress all safety considerations.

e. Gearing theory and data calculation instruction shall be given in the classroom.

10. Teaching Points.

a. Skills Practice: performed in concert with teaching points below.

b. Explain the various types of gear construction to include:

(1) spur gear;

(2) internal gears;

(3) helical gears;

(4) herring bone gears;

(5) bevel gears;

(6) rack and pinion gears; and

(7) worm gears.

c. Explain the terminology which applies to the gears physical construction to include:

(1) Addendum;

(2) centre distance;

(3) chordal addendum;

(4) chordal thickness;

(5) circular pitch;

(6) circular thickness;

(7) clearance;

(8) dedendum;

(9) diametral pitch;

(10) linear pitch;

(11) module;

(12) outside diameter;

(13) pitch circle;

(14) pitch circumference;

(15) pitch diameter;

(16) pressure angle;

(17) root angle/circle;

(18) root diameter;

(19) tooth thickness,

(20) whole depth; and

(21) working depth.

d. Explain the concept of Involute gear cutters, including their use and applications.

e. Explain and demonstrate the procedure for calculating gear data using the formulas.

f. Skills Practice: Explain and demonstrate the procedure for cutting a spur gear.

## PO 006 Perform Advanced Milling Machine Operation

## EO 006.05

1. Performance. Perform Advanced Milling Operations.

2. Conditions.

   a. Given:

      (1) milling machine,

      (2) machinist's tool box,

      (3) metal stock,

      (4) indexing head,

      (5) milling cutters,

      (6) drawings,

      (7) references,

      (8) assistance, and

      (9) supervision.

   b. Denied: n/a

   c. Environmental:

      (1) Equipped machine shop

3. Standard.

   a. Obey all safety procedures, see ref C5;

   b. Working to specific tolerances as per working drawings IAW ref D3:, mill hex and keyways, see ”Indexing" ref C4, C5, and V-block and clamp.

4. Time.

   a. 2 x 50 minute periods for instruction

   b. 28 x 50 minute periods for EC 006.05 (Hex and Keyways)

c. 56 x 50 minute periods for PC 006 (V-Block and clamp)

5. Method/Media.

a. Instruction

b. Video

c. Skills Practical

6. References.

C1, C3, C5, D3

7. Training Aids.

a. See para 2a.

b. Videos 38660 6345, 38660 6346, 38660 6349 and 38660 6350.

8. Test Details.

a. Practical EC 006.05: Hex and Keyway Project

b. Practical PC 006: V-Block and Clamp Project

c. Marked by means of checklist(s) IAW ref D3

d. Minimum pass mark 75%

e. 56 x 50 minute periods for PC 006

f. 28 x 50 minute periods for EC 006.05

9. Remarks.

a. The instructor shall set up all displays prior to commencing instruction.

b. The instructor will ensure that hands on" participation is started early in the lesson.

c. The instructor will verify the results of the hex and keyway and the V-block and clamp.

d. Trainees are required to read reference material before instruction.

e. Practical EC to be administered by instructor to confirm TP covered in EO

006.05, to be completed prior to commencing PC 006

f. Practical PC 006 to assess TPs covered in PO 006

10. Teaching Points.

1. Explain and demonstrate the operation and application of the standard indexing head attachment.
2. Demonstrate the method of direct indexing.
3. Demonstrate and explain the method of indexing using the formula: 40/N
4. Explain the use and operation of the index plate and sector arms.
5. Demonstrate and explain the method of set-up and cutting a keyway into round stock.

## Enabling Check 006.05 – Hexagon and Keyways Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop to include milling machine and attachment;
2. Machinist tool box;
3. Hold down devices
4. Various milling cutters,
5. Precision layout tools to include:
6. Precision measuring tools to include:
7. Metal Stock:
   a. 1 ¾" Round, Cold Rolled Mild Steel, 5 ¼" Long (1 per student).
8. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Enabling check is to confirm TPs covered in EO 006.05 by completing a Hexagon and Keyways Milling Project IAW ref D3 (working drawing and specified tolerances);
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 28 x 50 minute periods to complete EC.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with EC checklist.
7. Pass mark is 75% of total marks in marking guide.
8. Each trainee will be permitted a maximum of three restarts.
9. There is a 5% penalty for restarts.

10. Failure to achieve a passing mark of 75% for this EC will result in completion of Trainee Experiencing Difficulty form.

11. Failure to follow all shop safety regulations and procedures will result in a failure of this EC.

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Enabling Check is to confirm the TPs covered in EO 006.05 by milling Hexagon and Keyway Project IAW working drawings and specified tolerances.
2. You will be given:
   a. Milling Machine
   b. Hold down devices;
   c. Various cutters;
   d. Metal Stock: 2” Round, Hot Rolled Mild Steel, 19” long;
   e. Machinist tool box;
   f. Precision layout tools
   g. Shop fluids;
   h. Working drawing complete with EC checklist; and
   i. Supervision.
3. You will be denied assistance
4. You will have 28 x 50 minute periods to complete EC;
5. Minimum pass mark is 75% IAW marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. Failure to achieve a passing mark will result in completion of Trainee Experiencing Difficulty Form

9. Observe and adhere to machine shop safety procedures, failure to do so will result in

**PART III - ENABLING CHECK NOTES**

1. Ensure keyways cut in correct order as per drawing
2. Digital measuring tools and digital readout displays may be used at instructors' discretion
3. Vertical milling operations to be used
4. Individual student project: trainees are not to share milling machines or set-ups with other trainees

## Performance Check 006 - V-Block and Clamp Project

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop: milling machine including attachments;
2. Machinist tool box;
3. Hold down devices;
4. Various milling cutters;
5. Precision layout tools to include:
6. Precision measuring tools to include:
7. Metal Stock (1 each per trainee):
    a. 2 ¼” Square, Mild Steel, 2” Long;
    b. 3/16” Drill Rod, 2 ½” Long;
    c. 5/16” Drill Rod, 2 ½” Long;
    d. ¾” Round, Hot Rolled Mild Steel, 4” Long;
    e. 1/8” Drill Rod, ½” Long;
    f. ½” Round, Hot Rolled Mild Steel, ¾” Long;
    g. ¾” x 4” Mild Steel Flat Bar;
    h. ¾” Square, Mild Steel, 1” Long; and
    i. ¼-20 UNC Set Screws, 4 ea.
8. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees Advanced Milling Machine operating skills by machining a "V" Block and Clamp as per working drawing and specified tolerances IAW ref D3;

2. Assistance is not permitted.

3. Supervision is mandatory.
4. Time allocated is 39 x 50 minute periods to complete PC.
5. Part II lists what each trainee will be given.
6. Trainees to be familiar with PC checklist.
7. Pass mark is 75% of total marks in marking guide.
8. Each trainee will be permitted a maximum of three restarts.
9. There is a 5% penalty for restarts.
10. There will be no re-writes of this PC for this PO; failure to achieve a passing mark for this PC will result in a TRB.
11. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
12. PC 005 Vee Block and Clamp represents 10% of the final overall mark for Naval Machinist Course

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Performance Check is an evaluation of your milling machine operating abilities.
2. You will be given:
    a. Equipped machine shop: individual lathes and milling machines equipped with attachments
    b. Machinist tool box;
    c. Hold down devices;
    d. Various milling cutters;
    e. Precision layout tools;
    f. Precision measuring tools;
    g. Metal Stock,

h. Shop fluids; and

i. Supervision.

3. You will be denied assistance
4. You will have 39 x 50 minute periods to complete PC 006;
5. Minimum pass mark is 75% IAW PC 006 marking guide;
6. You will be permitted a maximum of three restarts;
7. There is a 5% penalty for restarts;
8. There will be no re-writes for PC 006; failure to achieve a passing mark will result in a TRB; and
9. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
10. PC 005 Vee Block and Clamp represents 10% of the final overall mark for Naval Machinist Course

**PART III - PERFORMANCE CHECK NOTES**

1. Both vertical and horizontal milling operations to be used at trainees' discretion.
2. Project must be complete and assembled for pass.
3. Digital measuring tools and Digital read-out displays used at instructors discretion.
4. All external thread forms to be manufactured on the lathe: die nuts or other aids are not permitted.

## PO 007 Machine Components

## EO 007.01

1. Performance. Fabricate Components

2. Conditions.

   a. Given:

      (1) Operational Machine Shop

      (2) references, and

      (3) supervision.

   b. Denied: Assistance

   c. Environmental:

      a. equipped machine workshop.

3. Standard.

   a. Fabricate Pump Shaft, Gear Pump and Drill Press projects IAW ref D3.

   b. Adhere to all safety and operating procedures, ref A1, C5.

4. Time.

   a. 191 x 50 Minute Periods for PC 007 Part A (Drill Press).

   b. 159 x 50 Minute periods for PC 007 Part B (Gear Pump).

   c. 82 x 50 Minute Periods for PC 007 Part C (Pump Shaft).

5. Method/Media.

   a. Practical

6. References.

   A1, A2, A3, A4, C1, C2, C3, C4, C5, D1, D2, D3

7. Training Aids.

   a. See para 2.

8. Test Details.

a. Practical PC 007: The PC for this PO will culminate skills and knowledge covered in all the POs into three projects; Part A Drill Press, Part B Gear Pump, and Part C Pump Shaft IAW ref D3.

b. Minimum pass mark for this PC 007 is 75%.

c. 432 minute periods for PC 007

9. Remarks.

a. PC 007 will confirm and assess skills and knowledge attained from POs 001 thru PO 006.

b. The instructor shall stress all safety considerations.

c. The Gear Pump must function IAW the specification as per Ref. D3,

d. The Gear Pump casing shall be machined using a lathe. The cover may be manufactured using the milling machine.

e. Each student is responsible for individual machine setups for each project.

f. Instructors are to ensure that using previous machine setups from other students is prohibited.

10. Teaching Points.

N/A

## Performance Check 007 Part A – Drill Press

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop;
2. Machinist tool box;
3. Machinist Hand Book;
4. Machinist tool;
5. Precision layout tools;
6. Precision measuring tools;
7. Metal Stock (1 each per trainee):
    a. 1" Round, Brass, 4” long;
    b. 1 1/2" Round, Brass, 3” Long;
    c. 1 1/8" Round, Brass, %” Long;
    d. 1 3/4” Round, Cold Rolled Mild Steel, 7” Long;
    e. 1/2" Drill Rod, 12” Long;
    f. 1/8" Square Wire, Mild Steel, 6” Long
    g. 2" Boiler Tube, 24 ¼” Long;
    h. 2" Round, Brass, 5 ¾” Long;
    i. 3/8" Drill Rod, 13” Long;
    j. 4 1/4" Round, 6061 T6 Aluminum, 8” Long;
    k. 5/16" Drill Rod, 3 ¼” Long;
    l. 5/8" Drill Rod, 10” Long;
    m. 7/8" Drill Rod, 3.35” Long;
    n. Bolt, 1/2-13 X 1" Long;

o. Bolt, 1/2-13 X 3" Long;

p. Bolt, 3/8-16 X 1&1/2" Long;

q. Casting, Base Plate;

r. Casting, Motor Support;

s. Casting, Spindle Support (head);

t. Casting, Table;

u. Casting, Table Bracket;

v. Chuck and key;

w. Set-screw, 1/4-20;

x. Set-screw, 10-24;

y. Set-screw, 8-32; and

z. Spring Return Assembly

8. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees Advanced Milling Machine and Lathe operating skills by manufacturing a Drill Press as per working drawing and specified tolerances IAW ref D3;
2. Review working drawing and PC marking guide;
3. Review set-up procedures concerning castings;
4. Emphasize important tolerances in relation to fits.
5. Spindle run-out to within tolerances.
6. Assistance is not permitted.
7. Supervision is mandatory.
8. Time allocated is 191 x 50 minute periods to complete Part A of PC 007.
9. Part II lists what each trainee will be given.

10. Pass mark is 75% of total marks IAW Part A (Drill Press) marking guide.

11. There is a 5% penalty for restarts;

12. Each trainee will be permitted a maximum of three restarts for entire PC (Part A, Part B and Part C inclusive).

13. There will be no re-writes for Part A of this PC: failure to achieve a passing mark for Part A of this PC could result in a TRB.

14. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.

15. Part A of PC 007 (Drill Press) represents 40 % of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Performance Check is an evaluation of your ability to plan and execute the various machining procedures you have been taught.

2. You will be given:

   a. Equipped machine shop;

   b. Machinist tool box;

   c. Machinist hand book;

   d. Machine tools;

   e. Precision layout tools:

   f. Precision measuring tools:

   g. Metal Stock,

   h. Shop fluids; and

   i. Supervision.

3. You will be denied assistance

4. You will have 191 x 50 minute periods to complete Part A of PC 007;

5. Minimum pass mark is 75% IAW Part A (Drill Press) marking guide;

6. There is a 5% penalty for restarts;

7. You will be permitted a maximum of three restarts for entire PC (Part A Part B and Part C inclusive);

8. There will be no re-writes for Part A of PC007; Failure to achieve a passing mark could result in a TRB.

9. Failure to follow all shop safety regulations and procedures will result in a failure of the PC.

10. Part A of PC 007 (Drill Press) represents 40 % of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART III - PERFORMANCE CHECK NOTES**

1. Both vertical and horizontal milling operations to be used at trainees' discretion.

2. Project must be complete and assembled for pass.

3. Digital measuring tools and Digital read-out displays used at instructors discretion.

4. All external thread forms to be manufactured on the lathe: die nuts or other aids are not permitted.

## Performance Check 007 Part B – Gear Pump

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop
2. Machinist tool box;
3. Machinist hand book;
4. Machine tools;
5. Precision layout tools;
6. Precision measuring tools;
7. Metal Stock (1 each per trainee):
    a. 5/8” Drill Rod, 14” Long;
    b. 1 ¼” Hex stock, Brass, 4” Long;
    c. 2 ½” Round, Brass stock, 3” Long;
    d. 1/8” Drill Rod, 4” Long;
    e. ¾” Drill Rod, 3 ¾” Long;
    f. 1” Drill Rod, 2 ½” Long; and
8. Gear Pump casing and Cover Castings
9. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees ability to machine component pieces from various metals and fit them to close tolerances as per working drawing and specified tolerances IAW ref D3;
2. The trainee will be given working drawings complete with Part B (Gear Pump) checklist before practical assessment.

3. The trainee is required to develop a suitable work procedure plan.

4. Full use of Machine Shop equipment is permitted.

5. Assistance is not permitted.

6. Supervision is mandatory.

7. Time allocated is 159 x 50 minute periods to complete Part B of PC007.

8. Part II lists what each trainee will be given.

9. Pass mark is 75% of total marks IAW Part B (Gear Pump) marking guide.

10. There is a 5% penalty for restarts.

11. Each trainee will be permitted a maximum of three restarts for PC 007 (Part A, Part B and Part C inclusive).

12. There will be no re-writes for Part B; failure to achieve a passing mark for Part B of this PC could result in a TRB.

13. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.

14. Part B of PC 007 (Gear Pump) represents 30 % of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Performance Check will assess your ability to carryout complex machining operations. You will be required to work to close tolerances using various types of material. A successful operational test is required in order to pass this Part B of this PC.

2. You will be given:

   c. Equipped machine shop;

   d. Machinist tool box;

e. Machine tools;

f. Machinist hand book;

g. Precision layout tools;

h. Precision measuring tools;

i. Metal Stock;

j. Shop fluids;

k. Working drawing complete with Part B (Gear Pump) checklist; and

l. Supervision.

3. You will be denied assistance
4. You will have 159 x 50 minute periods to complete this PC;
5. Minimum pass mark is 75% IAW Part B (Gear Pump) marking guide;
6. You will be permitted a maximum of three restarts for PC 007 (Part A, Part B and Part C inclusive);
7. There is a 5% penalty for restarts;
8. There will be no re-writes for Part B of PC 007; Failure to achieve a passing mark could result in a TRB.
9. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
10. Part B of PC 007 (Gear Pump) represents 30 % of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART III - PERFORMANCE CHECK NOTES**

1. Trainees to plan their own job set-ups using gear pump procedure sheet as a guide. Instructor to ensure that trainees learn how to chuck-up and dial in unusual job sets (Gear Pump Casing and Cover)

2. Digital measuring tools and Digital read-out displays used at instructors discretion.

3. This is an individual trainee PC project – trainees will not share machine or work piece set-ups with other trainees.
4. See “Pump Test Rig” drawing for correct test set-up
5. All pipe threads are national pipe taper. Tap these threads for standard 3 ¼ to 3 ½ turns from “two fingers” tight to disengagement.
6. Pump must be presented fully assembled on the test rig stand for a passing mark

## Performance Check 007 Part C – Pump Shaft

**PART I - ADMINISTRATIVE INSTRUCTIONS**

**MATERIAL AND FACILITIES REQUIRED**

1. Equipped machine shop;
2. Machinist tool box;
3. Machinist hand book;
4. Machine tools;
5. Precision layout tools;
6. Precision measuring tools;
7. Metal Stock (1 each per trainee):
    a. 2” Round, Brass Stock, 2 1/2” Long;
    b. 2” Round, Mild Steel, 19” Long;
    c. 2 ¼ Hex stock, Brass, 3” Long;
    d. 2 1/2 “ Round, Hot Rolled Mild Steel, 7/8” Long;
    e. 1 1/16” Hex stock, Stainless Steel, 7/8” Long;
    f. 3/8” Square Stock or Key stock, Mild Steel, 3” Long; and
    g. #807 Woodruff key.
8. Shop fluids

**ARRANGEMENTS AND PROCEDURES**

1. This Performance check is an evaluation of the trainees ability as a Naval Machinist to manufacture replacement parts to specific tolerances IAW working drawings;
2. Assistance is not permitted.
3. Supervision is mandatory.
4. Time allocated is 82 x 50 minute periods to complete Part C (Pump Shaft) of PC 007.

5. Part II lists what each trainee will be given.

6. Trainees to be familiar with Part C (Pump Shaft) checklist.

7. Pass mark is 75% of total marks IAW Part C (Pump Shaft) marking guide.

8. Each trainee will be permitted a maximum of three restarts for PC 007 (Part A, Part B and Part C inclusive).

9. There is a 5% penalty for restarts.

10. There will be no re-writes for Part C of PC 007; failure to achieve a passing mark for Part C could result in a TRB.

11. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.

12. Part C of PC 007 (Pump Shaft) represents 30% of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART II - INSTRUCTIONS GIVEN TO TRAINEE**

1. This Performance Check is an evaluation of your ability to combine lathe and milling machine procedures.

2. You will be given:

   a. Equipped machine shop: individual lathes and milling machines equipped with attachments;

   b. Machinist tool box;

   c. Machinist hand book;

   d. Machine tools;

   e. Precision layout tools;

   f. Precision measuring tools;

   g. Metal Stock;

h. Shop fluids; and

i. Supervision.

3. You will be denied assistance
4. You will have 82 x 50 minute periods to complete Part C (Pump Shaft) of PC 007;
5. Minimum pass mark is 75% IAW Part C (Pump Shaft) marking guide;
6. You will be permitted a maximum of three restarts for PC 007 (Part A, Part B and Part C inclusive);
7. There is a 5% penalty for restarts;
8. There will be no re-writes for Part C of PC 007; failure to achieve a passing mark could result in a TRB.
9. Failure to follow all shop safety regulations and procedures will result in a failure of this PC.
10. Part C of PC 007 (Pump Shaft) represents 30% of PC 007, PC 007 (Part A, Part B and Part C combined) represents 70% of the final overall mark for Naval Machinist Course

**PART III - PERFORMANCE CHECK NOTES**

1. Milling and lathe operations/set-ups to be at trainees' discretion.
2. Digital measuring tools and Digital read-out displays used at instructors discretion.
3. This is an individual trainee PC project – trainees will not share machine or work piece set-ups with other trainees.
4. All external thread forms shall be manufactured on the lathe- - Die nuts or other aids are not permitted
5. High Speed Tooling only
6. Project must be complete and assembled for pass.

# ANNEX A – Main References

| Reference Code | Publication | Quantity |
|---|---|---|
| A1 | C-02-040-009/AG-001 General Safety Program Vol. 1, General Safety Standards | 1 |
| A2 | MARCORD 66-5 WHMIS for Supervisors | 1 |
| A3 | C-02-040-009/AG-001 General Safety Program Vol. 4, Hazardous Material Safety | 1 |
| A4 | Canada Labour Code PT II | 1 |
| C1 | A. Krar, S.F. Oswald, J.W., Machine Shop Training | 8 |
| C2 | B. Walker, J.R., Modern Metalworking | 8 |
| C3 | C. Firer, J.L., Machine Tool Metalworking | 8 |
| C4 | D. Erik Oberg, Franklin D. Jone, Holbrook L. Horton, Machinery's Handbook, current edition | 8 |
| C5 | E. Krar, S.F., Oswald J.W., Technology of Machine Tools | 8 |
| D1 | Machinist Specialist Course Notes (handout) | 8 |
| D2 | Turnvision ACU-RITE Digital Readout Manual | 8 |
| D3 | Naval Machinist Course Specification Drawings | 1 |

# ANNEX B –Task List

| **Task** Serial | **TASK ELEMENT** | **Trg / No Trg** | **PO** | **Justification (No Train)** |
|---|---|---|---|---|
| 1 | Enforce Shop Safety | Trg | 001 | |
| 2 | Evaluate Machinery | Trg | 001 | |
| 3 | Maintain Machine Shop Inventory | Trg | 001 | |
| 4 | Interpret Machining Procedure Specification | Trg | 001 | |
| 5 | Perform Precision Measuring | Trg | 002 | |
| 6 | Perform Precision Layout | Trg | 002 | |
| 7 | Perform Hand Fitting Operations | Trg | 002 | |
| 8 | Perform Advanced Grinding Operations | Trg | 003 | |
| 9 | Perform Advanced Machine Lathe Operations | Trg | 005 | |
| 10 | Perform Precision Drill Press Operations | Trg | 004 | |
| 11 | Perform Advanced Milling Machine Operations | Trg | 006 | |

# ANNEX C – RECORD OF PROCEEDINGS

Memorandum

4975-1

06 Oct 2004

DMTE 2

RECORD OF PROCEEDINGS (ROP)
QUALIFICATION STANDARDS AND PLAN (QSP)
NAVAL MACHINIST COURSE

Refs: A. Convening Order –Qualification Standard and Plan (QSP) Board, DMTE 1351, 181101Z Aug 04
B. A-P9-050 CF Manual of Individual Training and Education

QSP BOARD

1. A QSP Board was convened for the period 22 Sep – 07 Oct 04 at CFFS Esquimalt, IAW ref A. The board's mandates were to:

a. review and update the OSS AHHD;

b. review and update the QSP Naval Machinist IAW the reviewed OSS;

c. develop and input new Chap 1 and 2 of the Naval Machinist QSP; and

d. time permitting, develop and input Chap 3 and 4 of the Naval Machinist QSP.

2. This ROP will cover both the OSS review and QSP process.

Board Members:

| | | | |
|---|---|---|---|
| Chairperson | Lt(N) Jollimore | MS ENG 88 | MARPAC F4 ENG |
| TDO | Lt(N) Robinson | 74A | CFFS Esquimalt, Trng Dev |
| DMTE Rep | CPO1 Boucher | 00125 | DMTE 2-5-2 |
| Member | CPO2 Thomson | 00123 | CFFS Esquimalt/Shop CPO |
| Member | CPO2 Major | 00123 | CFFS Esquimalt/Standards |
| Member | CPO2 Cowden | 00123 | CFFS Esquimalt/Standards |
| Member | PO1 Pryor | 00123 | CFNES Halifax/Shop I/C/Instr |
| Member | PO1 Anderson | 00123 | CFFS Esquimalt/Shop I/C/Instr |
| Member | PO2 Taylor | 00123 | FMF CB Diesel Inspector |

| | | | |
|---|---|---|---|
| Member | PO2 Penner | 00123 | HMCS Protecteur |
| Member | PO2 Beaulac | 00123 | CFFS Esquimalt/Instructor |

AIM

3. The QSP Board mandate was to develop and submit revised OSS and QSP for the Naval Machinist Course.

NAVAL MACHINIST OSS AHHD

4. The board reviewed the OSS AHHD last modified 31 March 1992. This OSS was based on the old 3-point scaled vice the new 5-point scale. In reviewing the OSS, it was determined that the wording for some of the tasks should be changed to "advanced" since it more accurately reflects the scope of this specialty course.

5. The waiver process was discussed and this board determined that any MAR ENG QL5A qualified member wishing to become a naval machinist but lacking some of the required pre-requisites from their prior training, such as 80% in their machinist shop phase or the recommendation of their QL5A course instructor may apply to their respective fleet school to perform a waiver project when given the recommendation of their CERA. The respective fleet school under the supervision of the shop I/C will then assign and mark the waiver project. If their waiver projects final mark is 80% or higher, the Commandant of CFFSE/CFNES will be approached in order to grant waiver. The waiver project has been agreed by all board members and has been included in the Naval Machinist QSP along with a marking guide.

6. Heat treatment process, understanding digital readout and casting procedures already covered in the Naval Machinist course have been included in the OSS. The knowledge "casting procedures" was discussed in great length and will be discussed in more details in the Naval Machinist QSP.

7. It is the board's recommendation that the above OSS documents be accepted as revised. As defined in this OSS documents, CFFS Esquimalt and CFNES Halifax will handle all waiver request for their respective coasts using the standardized project and marking scheme to maintain parity with each other.

1. NAVAL MACHINIST QSP

8. The board then examined the Naval Machinist QSP. It was decided to increase the number of Performance Objectives (POs) to seven from five. The PO "Use Hand Tools" has been split in two different POs (PO 001 Manage a Machine Shop & PO 002 Perform Machine Shop Fitting). These two new POs have been created to reflect the actual position of the naval machinist onboard ship and to provide better flow when teaching this course.

9. Heat treatment process and casting procedures have been added to PO 001 (Manage a Machine Shop) while understanding digital readout has been added to PO 005

(Perform Advanced Machine Lathe Operations). CFFS Esquimalt's machine shop has six lathes and six milling machines fitted with digital readout while CFNES Halifax's machine shop has one lathe and one milling machine fitted with digital readout. Since the objective of this teaching point is to provide members with an understanding of digital readout, it is expected that one lathe and one milling machine will be sufficient to teach this topic. Heat treatment process and understanding digital readout that have been added to this QSP will not change the overall length of the course. An offset has been created by reducing the amount of time required to fabricate some of the projects.

10. The knowledge casting procedures has been discussed at great length by all board members. Although all agreed that two days at the foundry seemed excessive, the board was unanimous in favor of keeping this training the way it was. It was determined that the experience acquired during this phase was critical to a Naval Machinist Specialists understanding of the properties of cast materials and how these unique properties will affect the machinists work.

11. The existing QSP, had part PCs that were contradictory with CFITES policies. To correct this deficiency, the board decided to make PO 007, Machine Components. Here the part PCs in the old QSP where replaced with PO 007 where the final projects that encompass all the training received on this course will be evaluated.

12. The Naval Machinist course is considered by the board to be an elite course that offers advanced training to selected QL5A graduates. The nature of the job demands excellence so that parts fabricated under operational conditions perform the intended function and meet the restrictions of shipboard facilities as well as time. Therefore, the board recommends that the candidate must attain a minimum of 75% for each assessment (PC) throughout the course. It should be noted that the board was not unanimous in this opinion however a strong majority carried the vote in favor of raising the minimum acceptable standard for this specialist course.

13. <u>Chapters 1 and 2.</u> They have been reviewed and amended IAW paragraphs above using Microsoft Word. NTDSS was not available at the time of this QSPWB. As indicated by the DMTE Rep, Chapters 1 and 2 and all annexes will be entered in NTDSS by DMTE.

14. <u>Chapter 3 and 4.</u> All PO/EOs have been reviewed and amended for performance, conditions, standard, method/media, references, training aids, remarks and teaching points. This QSPWB did not have sufficient time to complete all the requirement of Chapter 4 IAW NTDSS QSP. Times and test details have not been included in Chapters 3 and 4 as they are to be promulgated by CFFS Esquimalt Standard Rep. All board members agreed to have Chapters 3 & 4 Times and Test Details to be completed and entered in NTDSS by CFFS Esquimalt Standards.

15. <u>QSP Annex A - References.</u> The General Safety Program, Canada Labour Code, MARCORD 66-5, Digital Readout Manual, Naval Machinist Handout and Naval Machine Specification Drawings have been added. The Naval Machinist Handout has been created by CFFS Esquimalt and is issued to the students to keep. CFNES Halifax

has been given a copy and will reproduce locally. Since drawings could not be included in NTDSS, a Naval Machinist Specification Drawings reference was created that includes all projects that will be performed during the Naval Machinist course.

16. QSP Annex B – Task List. All tasks listed in the revised OSS will be trained and have been identified for each PO.

17. QSP Annex D – Qualification Standard and Plan Writing Board Scalar. Since the board decided to break-up and create POs, a scalar was developed to document the content and structure of the course.

18. QSP Annex E – On Job Performance Record. There is no requirement for an OJPR for the Naval Machinist course. Upon successful completion of this course, the student can be employed as Naval Machinist and be awarded the qualification AHHD.

19. QSP Annex F – Acronyms/Abbreviations. None identified.

20. It is the board's recommendation that the above course document be accepted as revised.

C.B. Jollimore, Lt(N)
Board Chairman

Annexes:

Annex A Occupation Specialty Specification – Naval Machinist Qualification AHHD

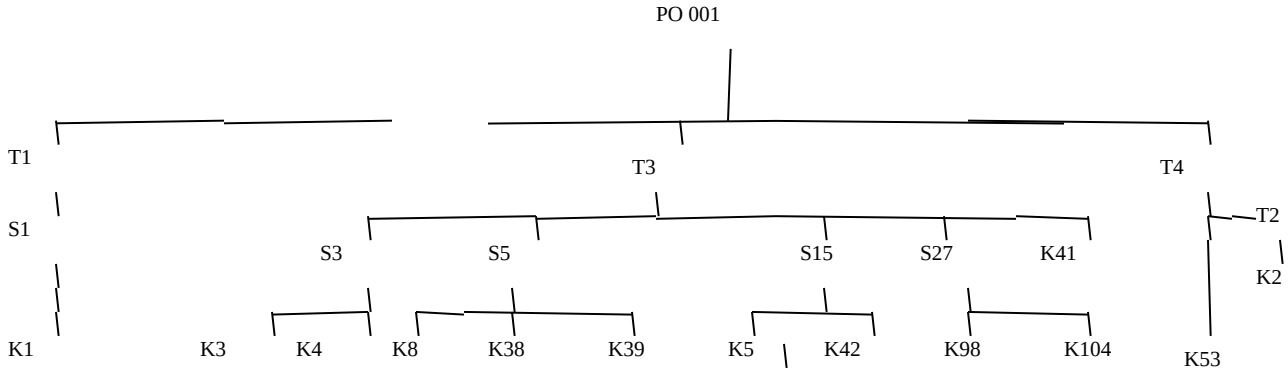
Annex B Quality Standards and Plan – Chap 1 & 2, Naval Machinist Course

Annex C References

Annex D Task List

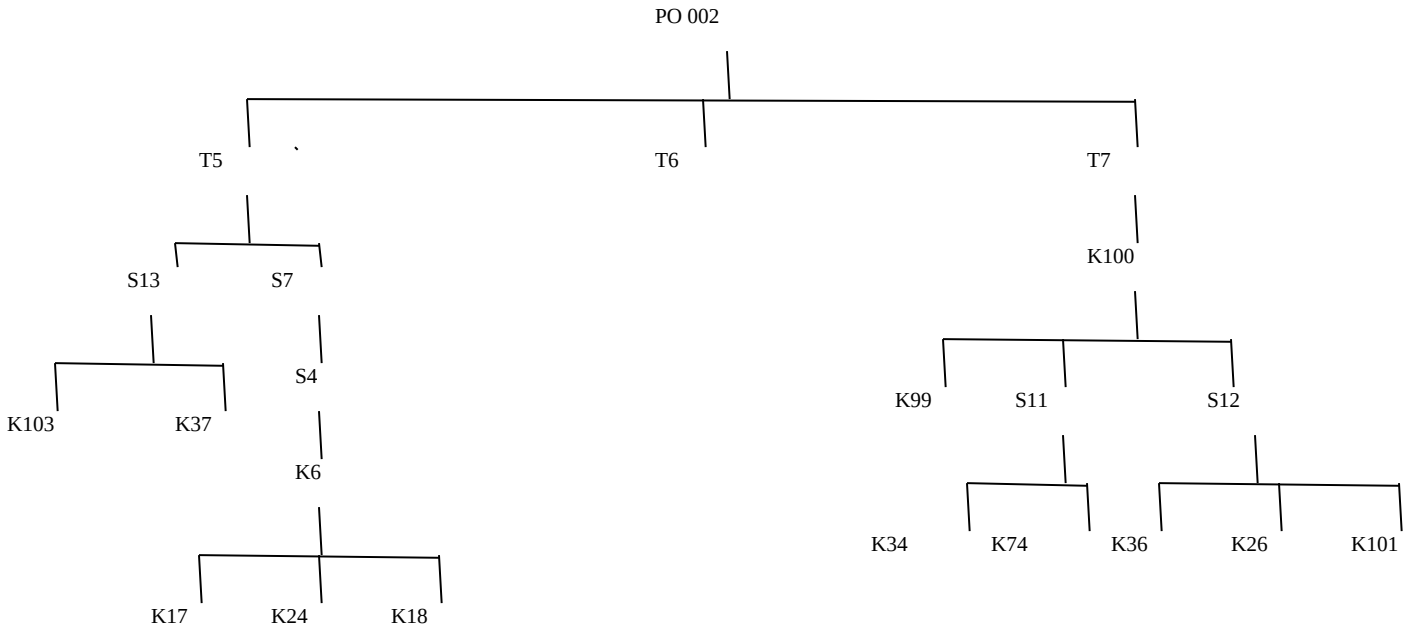
Annex E Qualification Standard and Plan Writing Board Scalar

# ANNEX D – QUALIFICATION STANDARD AND PLAN SCALAR



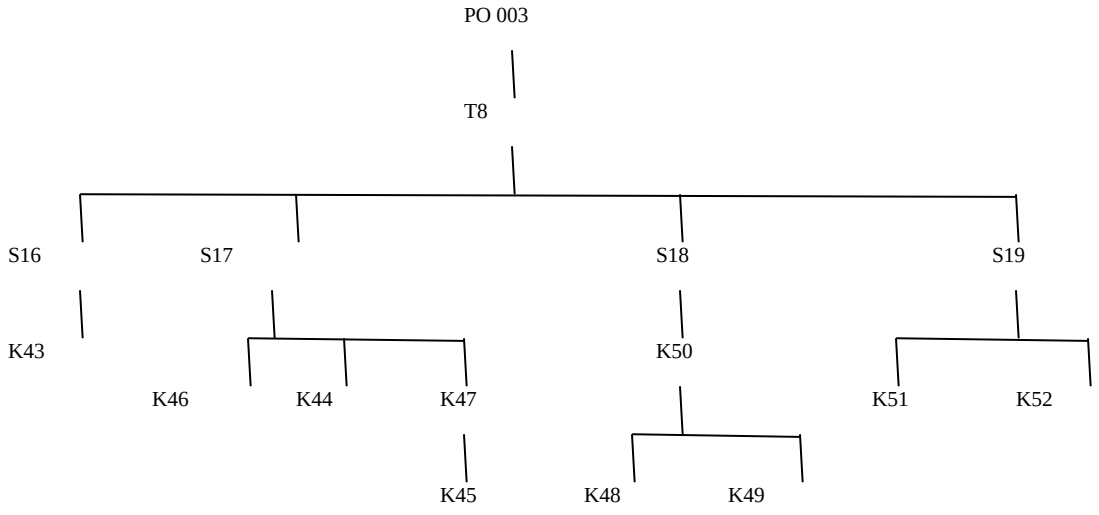

LEGEND PO 001

T1 Enforce Shop Safety
T2 Evaluate machinery
T3 Maintain Machine Shop Inventory
T4 Interpret Machining Procedure Specifications
S1 Identify Unsafe Shop Procedures
S3 Identify Metals
S5 Selecting Cutting Fluids
S15 Storing Tools
S27 Heat Treatment of Various Metals
K1 Safety Regulations and Procedures
K2 Working Parts of Shop Equipment
K3 AISI and SAE Codes
K4 Properties of Metals
K5 Machine Shop Cutting Tools
K8 Types of Cutting Fluids
K38 Solvent Types
K39 Lubricant Types
K41 Material Storage Considerations
K42 Tool Storage Considerations
K53 Machine Shop Calculations
K98 Heat Treatment Processes
K104 Casting Procedures



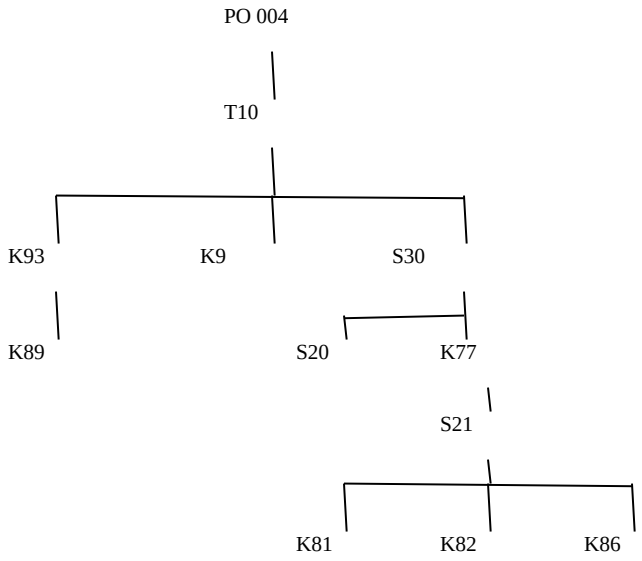

LEGEND PO 002

T5 Perform Precision Measurement
T6 Perform Precision Layout
T7 Perform Hand Fitting Operations
S4 Identifying Tools
S7 Select and Apply Measuring Tools
S11 Use Hand Cutting Tools
S12 Use Hand Finishing Tools
S13 Determine the Accuracy of Measuring Tools
K6 Measuring tools
K17 Vernier Measuring Instruments
K18 Micrometer Measuring Instruments
K24 Precision Measuring Tools
K26 Lapping Plates
K34 Application of Reamers
K36 Selection of Grinding and Lapping Compounds
K37 Micrometer Standards
K74 Broaching Keyways
K99 Classes of Fit
K100 Precision Fitting
K101 Surface Finishing
K103 Gauge blocks and Accessories



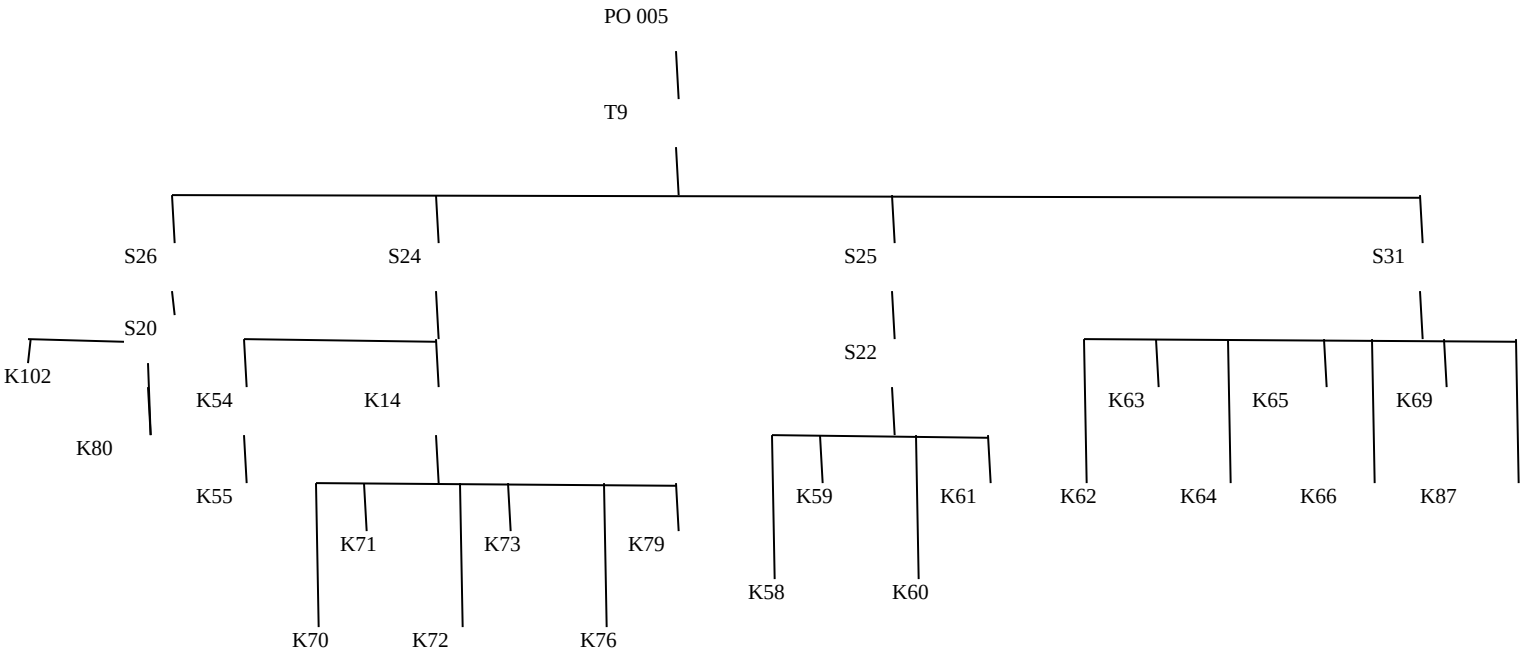

LEGEND PO 003

T8 Perform Advanced Grinding Operations
S16 Identify and Select Abrasive Wheels
S17 Testing and Mounting of Abrasive Wheels
S18 Truing and Dressing of Abrasive Wheels
S19 Grinding Techniques
K43 Classification of Grinding Wheels
K44 Grinding Wheel Speed Ranges
K45 Grinding Wheel Torque Limits
K46 Grinding Wheel Test Methods
K47 Grinding Wheel Mounting Procedures
K48 Grinding Wheel Dressing Tools
K49 Grinding Wheel Truing Tools
K50 Truing and Dressing Procedures
K51 Grinding Tool Bits
K52 Sharpening Drill Bits



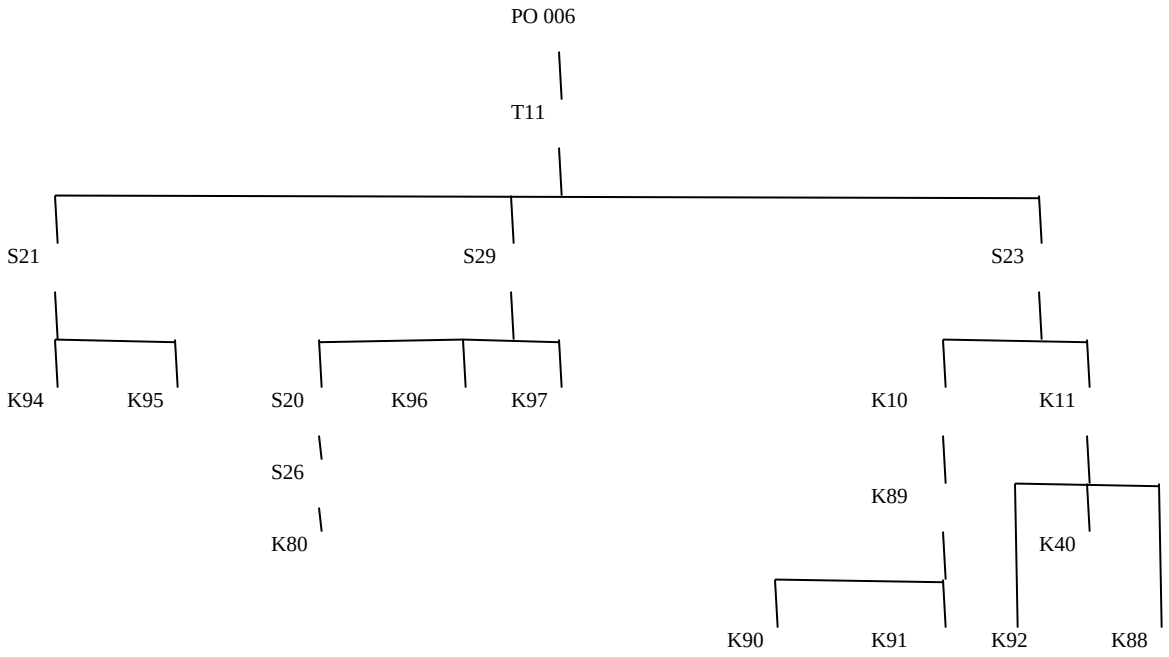

LEGEND PO 004

T10 Perform Precision Drill Press Operations
S20 Calculation of Spindle Speeds and Feed Rates (Drill Press)
S21 Using Cutting Tools and Specialty Cutters (Drill Press)
S30 Precision Drill Press Operation
K9 Application of Various Cutting Fluids
K77 Drilling Operations
K81 Step Drill
K82 Spot Facers
K86 Fly Cutters
K89 Work Holding Devices
K93 Clamps and Fasteners



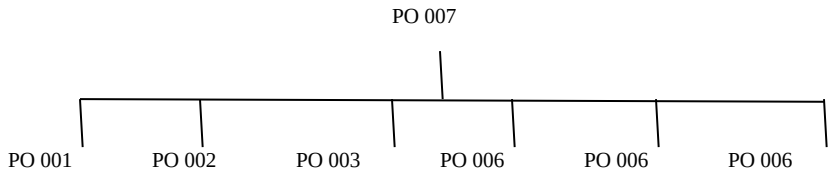

## LEGEND PO 005

T9 Perform Advanced Machine Lathe Operation
S20 Calculation of Spindle speed and Feed Rates
S22 Aligning Tailstock
S24 Precision Lathe Operation
S25 Alignment and Setup Procedures
S26 Calculating Cutting Speeds and Rates of Feed
S31 Using Lathe Attachments
K14 Machining Techniques
K54 Types and Applications of Lathe Cutting Tools
K55 Threading Tools
K58 Mandrel Alignment Method
K59 Centre-to-Centre Alignment Method
K60 Trial Cut Alignment Method
K61 Index Marks Alignment Method
K62 Application of Lathe Chucks
K63 Application of Drive Plates
K64 Application of Face Plates
K65 Follower Rest
K66 Steady Rest
K69 Tool Post Grinder
K70 Threading Operations
K71 Boring Operations
K72 Power Reaming Operations
K73 Knurling Operations
K76 Parting Operations
K79 Cutting Tapers
K80 Interpret Feed and Speed Tables
K87 Collets
K102 Understanding Digital Readouts




LEGEND PO 006

T11 Perform Advanced Milling Operations
S20 Calculation of Spindle Speeds and Feed Rates
S21 Using Cutting Tools and Specialty Cutters
S23 Using Milling Machine Attachments
S26 Calculate Cutting Speeds and Rates of Feed
S29 Precision milling Machine Operation
K10 Project Setups
K11 Machine Setups
K40 Assembly and Disassembly of Machine Tooling
K80 Interpreting Feed and Speed Tables
K88 Cutter Arbours and Adapters
K89 Work Holding Devices
K90 Rotary Table
K91 Indexing Head
K92 Vertical Milling Attachments
K94 Types and Applications of Milling Cutting Tools
K95 Handling of Cutters
K96 Procedures for Aligning of Cutters and Work pieces
K97 Procedures for Setting the Cutter to the Work Surface




PO 007 is a Culmination of the Knowledge acquired and consists entirely of the Pump Shaft, Gear Pump and Drill Press Projects.

# ANNEX H – Training Requirements

1. Classroom for 6 trainees to include:

   a. desks and chairs;
   b. whiteboard/flipchart;
   c. VCR and monitor;
   d. LCD projector; and
   e. PC with visual aids presentation software.

2. Training aids:

   (1) Machinist Tool Box (6/1 per student) each equipped with:

   - Callipers, Inside;
   - Callipers, Outside;
   - Dividers;
   - Callipers, Hermaphrodite;
   - Callipers, Vernier, 6”;
   - Scriber;
   - Machinists Square;
   - 6” Steel Rule;
   - 12” Steel Rule;
   - Centre Gauge and Holder, SAE;
   - Protractor Depth Gauge;
   - Derck’s Gauge and Simplifier;
   - Drill Point Gauge;
   - Telescopic Gauge Set (6);
   - Screw Pitch Gauge,SAE;
   - Screw Pitch Gauge, Metric;
   - Acme Thread Gauge;
   - Radius Gauge, large;
   - Radius Gauge, Small
   - Feeler Gauge;
   - Drill Drift;
   - Combination Square Set;
   - Mechanics’ style rolling Tool Chest;
   - Bubble Centre;
   - 6” Wrench, Adjustable;
   - 12” Wrench, Adjustable;
   - Standard 4” Screw Driver;
   - Standard 10” Screw Driver;
   - 8” Slip Joint Pliers;
   - Hacksaw;
   - Hammer, Ball Peen, 16oz;
   - Hammer Multi Tip (lead/copper);

- Hammer Dead Blow;
- File Card;
- File Handles (3);
- 12 Files with Cloth Pouch;
- 1 Set Jewellers Files;
- Prick Punch;
- Centre Punch;
- Centre Drills (3);
- Fractional drill Index with 29 Bits;
- Centre Gauge, Metric;
- Small Ruler Set W/Handle;
- Vise Grips;
- Allen wrench Set (metric);
- Allen Key set (SAE);
- Tin Snips (12 inch straight);
- Tin Snips (6 inch straight);
- Drive Pin Punch Set;
- Cold Chisels (2)
- Magnifying Glass;
- Edge/Center Finder (Wiggler);
- Magnetic Base;
- 0-1” micrometer;
- 1-2” micrometer;
- 2-3” micrometer;
- 3-4” micrometer;
- Dial indicator set;
- Depth Micrometer;
- Combination Wrench Set; and
- 3-wire gauge set;
- Honing stone;
- Lathe Cleaning Brush;
- Screw Jack;
- Wire Brush; and
- Scientific Calculator.

(2). PPE (aprons, safety glasses, ear protection, gloves…etc)

(3) Machine Shop equipped with:

a) Meets specifications of Colchester “Triumph” 2500 type Lathe or equivalent (minimum of 6 units 1 per student) to include:

- Lathe milling Attachment (min of 1);
- Lathe Rotary Table (min of 1);
- ACU-RITE Turnvision display unit or equivalent;
- Tool Post Grinder (min of 1);

- Dog Driving Plate;
- 3 Jaw Chuck and Key;
- Reversable Jaws (3 Jaw);
- 4 Jaw Chuck and key;
- Jacobs Chuck and Key;
- Collet Chuck (33)
- 35 assorted collets
- Steady Rest
- Follower rest
- Taper Attachment
- Micrometer Stop
- Coolant Pump System
- Live Centre
- Live Dead Centre
- Dead Centre;
- ½ Dead cetre;
- American Standard (AS) Tool post
- AS tool holders (Straight, Left Hand and Right Hand Offset) Offset 3/8" HSS tool bit (16 ½ degree back rake);
- AS Tool Holders (straight, RH, LH) for 3/8 Carbide Tool Bit;
- Knurling Tool
- Parting tool Holder, HSS
- Boring Bar Holder
- Assorted Boring Bars, HSS
- Indexing Turret
- Quick-change tool post with insert tooling
- Assorted Lathe Dog sizes (Standard bent tail, Straight tail and Safety clamp)

b) Knee and Column type milling machine (minimum of 6 units, (1 per student) or equivilant to include:

- both plain horizontal and vertical milling operations;
- Table feed and rapid traverse capability in X, Y, and Z axis;
- Universal table, minimum table size 12" X 48";
- Minimum power requirement 5HP with an arbour speed range of 50 to 1400 RPM;
- Micrometer Boring Head attachment in both horizontal and vertical positions;

- Universal Spiral Dividing (index) Head, complete with all attachments and index plates, capable of Direct and Simple indexing;
  - Machinist Vice;
  - Swivel Vice;
  - 18" Rotary Table;
  - Micrometer Boring Heads (minimum 3);

- Assorted (Jack Blocks, Vee Blocks'Step, Blocks, Angle plates, Parallel Bars, Clamp plates); and
- Arbours and spindle cutter adaptors for plain and vertical milling cutters for those cutters used in shop.

c). Pedestal type grinders (6);

d) Drill press (2) equipped with auto feed, vice, c-clamps, machinist clamps, and drill chuck;

e) Band saw;

f) Cut-off saw;

g) Hydraulic press;

h) Metal stock; and

i) Tool crib with specialty tools and equipment.

(4) Fitting shop;

(5) Technical library; and

(6) Videos: (insert video ref numbers)

| **Video #** | **Title** | 2. Time |
|---|---|---|
| 38657 6301 | Safety Procedures and Guidelines | 11 min |
| 38657 6302 | Hand tools and their use | 11 min |
| 38657 6303 | The use of measuring tools | 11 min |
| 38657 6304 | Micrometers: calipers, telescopic gauges, hole gauges | 18 min |
| 38657 6305 | Vernier scale; vernier, inside, outside and dial calipers | 13 min |
| 38657 6306 | The use of layout tools | 17 min |
| 38657 6307 | The pedestal grinder | 8 min |
| 38657 6308 | Sharpening drill bits | 12 min |
| 38657 6309 | Drill presses – sensitive and radical | 8 min |
| 38657 6310 | Operations frequently performed on the drill presses | 12 min |
| 38657 6311 | Hacksaw and filing procedures | 14 min |
| 38657 6312 | Vertical band machine parts and accessories | 10 min |
| 38657 6313 | Butt welding saw blades | 10 min |
| 38657 6314 | Band machine set up and use of accessories | 14 min |
| 38658 6315 | Identification of parts and care of the engine lathe | 13 min |
| 38658 6316 | Engine lathe accessories | 13 min |
| 38658 6317 | Cutting speeds and feeds for the engine lathe | 16 min |
| 38658 6318 | Grinding a right hand roughing tool | 13 min |
| 38658 6319 | Grinding a round nose finishing tool | 11 min |
| 38658 6320 | Mounting and truing in a 4 jaw independent chuck | 14 min |
| 38658 6321 | Three methods of facing work to length | 20 min |
| 38658 6322 | Straight turning work of two diameters | 13 min |
| 38658 6323 | Straight turning between centers | 17 min |

| | | |
|---|---|---|
| 38658 6324 | Drilling, boring & reaming work held in a lathe chuck | 20 min |
| 38658 6325 | Turning a radius | 16 min |
| 38658 6326 | Taper turning on a lathe | 20 min |
| 38658 6327 | Filing and polishing on the engine lathe | 13 min |
| 38658 6328 | Knurling on the lathe | 16 min |
| 38659 6329 | Grinding forming tools | 20 min |
| 38659 6330 | Machining shoulders and corners on the lathe | 16 min |
| 38659 6331 | Using the cut off tool on the lathe | 12 min |
| 38659 6332 | Measuring external and internal threads | 16 min |
| 38659 6333 | Tap and die threading on the lathe | 14 min |
| 38659 6334 | Chasing external threads and finishing with a die | 12 min |
| 38659 6335 | Roughing and finishing external threads | 15 min |
| 38659 6336 | Cutting internal threads on the lathe | 10 min |
| 38659 6337 | Cutting ACME threads | 15 min |
| 38659 6338 | Chasing tapered pipe threads on the lathe | 10 min |
| 38659 6339 | Machining offset holes – face plate & 4 jaw chuck | 18 min |
| 38659 6340 | Using the tool post grinder on a lathe | 11 min |
| 38659 6341 | Using the steady rest and follower rest to machine on the lathe | 9 min |
| 38659 6342 | Spring winding on the lathe | 8 min |
| 38660 6343 | Parts & operation of the vertical milling machine | 17 min |
| 38660 6344 | Cutters & machining operations for vertical milling machines | 18 min |
| 38660 6345 | Use of dividing head and rotary table | 12 min |
| 38660 6346 | Keyway on vertical milling machine: woodruff and straight | 13 min |
| 38660 6347 | Use of plain and side milling cutters on horizontal machine | 12 min |
| 38660 6348 | Parts and operation of horizontal milling machine | 9 min |
| 38660 6349 | Set up for holding work to be milled | 7 min |
| 38660 6350 | Use of face milling cutter on the horizontal mill | 9 min |
| 38660 6351 | Sharpening the periphery of an end mill | 14 min |
| 38660 6352 | Sharpening ends of end mills | 13 min |
| 38660 6353 | Gouging of 2 & 4 flute end mills | 11 min |
| 38660 6354 | Sharpening a reamer between centers | 11 min |
| 38660 6355 | Sharpening periphery plain milling cutters & side mill | 10 min |
| 38660 6356 | Sharpening brayed carbide lathe tools | 19 min |